

# US Patent & Trademark Office

## Patent Public Search | Text View

---

|                      |                      |
|----------------------|----------------------|
| United States Patent | 12408869             |
| Kind Code            | B2                   |
| Date of Patent       | September 09, 2025   |
| Inventor(s)          | Al-Ali; Ammar et al. |

---

### Disposable components for reusable physiological sensor

---

#### Abstract

A sensor cartridge according to embodiments of the disclosure is capable of being used with a non-invasive physiological sensor. Certain embodiments of the sensor cartridge protect the sensor from damage, such as damage due to repeated use, reduce the need for sensor sanitization, or both. Further, embodiments of the sensor cartridge are positionable on the user before insertion in the sensor and allow for improved alignment of the treatment site with the sensor. In addition, the sensor cartridge of certain embodiments of the disclosure can be configured to allow a single sensor to comfortably accommodate treatment sites of various sizes such as for both adult and pediatric applications.

---

**Inventors:** Al-Ali; Ammar (San Juan Capistrano, CA), Lamego; Marcelo (Cupertino, CA), Litchfield; Jim (Singapore, SG), Olsen; Gregory A. (Lake Forest, CA)

**Applicant:** Masimo Corporation (Irvine, CA)

**Family ID:** 42371531

**Assignee:** Masimo Corporation (Irvine, CA)

**Appl. No.:** 17/656982

**Filed:** March 29, 2022

#### Prior Publication Data

| Document Identifier | Publication Date |
|---------------------|------------------|
| US 20220218284 A1   | Jul. 14, 2022    |

#### Related U.S. Application Data

continuation parent-doc US 16425592 20190529 US 11331042 child-doc US 17656982  
continuation parent-doc US 15864389 20180108 US 10342487 20190709 child-doc US 16425592  
continuation parent-doc US 14626570 20150219 US 9895107 20180220 child-doc US 15864389

## Publication Classification

**Int. Cl.:** **A61B5/1455** (20060101); **A61B5/00** (20060101); **A61B5/0205** (20060101); **A61B5/024** (20060101); **A61B46/10** (20160101); **A61B46/20** (20160101)

**U.S. Cl.:**

**CPC** **A61B5/6838** (20130101); **A61B5/0205** (20130101); **A61B5/02444** (20130101); **A61B5/14552** (20130101); **A61B5/6826** (20130101); **A61B46/10** (20160201); **A61B46/20** (20160201); A61B5/024 (20130101); A61B2046/201 (20160201); A61B2560/0285 (20130101); A61B2562/247 (20130101)

## Field of Classification Search

**USPC:** None

---

## References Cited

### U.S. PATENT DOCUMENTS

| Patent No. | Issued Date | Patentee Name       | U.S. Cl. | CPC |
|------------|-------------|---------------------|----------|-----|
| 3463142    | 12/1968     | Harte               | N/A      | N/A |
| 3647299    | 12/1971     | Lavallee            | N/A      | N/A |
| 3740570    | 12/1972     | Kaelin et al.       | N/A      | N/A |
| 3799672    | 12/1973     | Vurek               | N/A      | N/A |
| 4086915    | 12/1977     | Kofsky et al.       | N/A      | N/A |
| 4169976    | 12/1978     | Cirri               | N/A      | N/A |
| 4182977    | 12/1979     | Stricklin, Jr.      | N/A      | N/A |
| 4308456    | 12/1980     | Van Der Gaag et al. | N/A      | N/A |
| 4346590    | 12/1981     | Brown               | N/A      | N/A |
| 4407290    | 12/1982     | Wilber              | N/A      | N/A |
| 4449821    | 12/1983     | Lee                 | N/A      | N/A |
| 4480886    | 12/1983     | Bergamin            | N/A      | N/A |
| 4580867    | 12/1985     | Wright et al.       | N/A      | N/A |
| 4621643    | 12/1985     | New, Jr. et al.     | N/A      | N/A |
| 4653498    | 12/1986     | New, Jr. et al.     | N/A      | N/A |
| 4685464    | 12/1986     | Goldberger et al.   | N/A      | N/A |
| 4700708    | 12/1986     | New, Jr. et al.     | N/A      | N/A |
| 4770179    | 12/1987     | New, Jr. et al.     | N/A      | N/A |
| 4825872    | 12/1988     | Tan                 | N/A      | N/A |
| 4830014    | 12/1988     | Goodman et al.      | N/A      | N/A |
| 4848901    | 12/1988     | Hood, Jr.           | N/A      | N/A |
| 4865038    | 12/1988     | Rich et al.         | N/A      | N/A |
| 4877322    | 12/1988     | Hill                | N/A      | N/A |
| 4913150    | 12/1989     | Cheung et al.       | N/A      | N/A |
| 4942877    | 12/1989     | Sakai et al.        | N/A      | N/A |
| 4960128    | 12/1989     | Gordon et al.       | N/A      | N/A |

|         |         |                   |     |     |
|---------|---------|-------------------|-----|-----|
| 4964408 | 12/1989 | Hink et al.       | N/A | N/A |
| 4974591 | 12/1989 | Awazu et al.      | N/A | N/A |
| 5041187 | 12/1990 | Hink et al.       | N/A | N/A |
| 5058588 | 12/1990 | Kaestle           | N/A | N/A |
| 5069213 | 12/1990 | Hink et al.       | N/A | N/A |
| 5090410 | 12/1991 | Saper et al.      | N/A | N/A |
| 5094240 | 12/1991 | Muz               | N/A | N/A |
| 5113862 | 12/1991 | Mortazavi         | N/A | N/A |
| 5140228 | 12/1991 | Biegel            | N/A | N/A |
| 5158323 | 12/1991 | Yamamoto et al.   | N/A | N/A |
| 5163438 | 12/1991 | Gordon et al.     | N/A | N/A |
| 5170786 | 12/1991 | Thomas et al.     | N/A | N/A |
| 5209230 | 12/1992 | Swedlow et al.    | N/A | N/A |
| 5226417 | 12/1992 | Swedlow et al.    | N/A | N/A |
| 5246003 | 12/1992 | DeLonzor          | N/A | N/A |
| 5249576 | 12/1992 | Goldberger et al. | N/A | N/A |
| 5267562 | 12/1992 | Ukawa et al.      | N/A | N/A |
| 5273041 | 12/1992 | Richards et al.   | N/A | N/A |
| 5279295 | 12/1993 | Martens et al.    | N/A | N/A |
| 5287853 | 12/1993 | Vester et al.     | N/A | N/A |
| 5308919 | 12/1993 | Minnich           | N/A | N/A |
| 5311865 | 12/1993 | Mayeux            | N/A | N/A |
| 5319355 | 12/1993 | Russek            | N/A | N/A |
| 5337744 | 12/1993 | Branigan          | N/A | N/A |
| 5339810 | 12/1993 | Ivers et al.      | N/A | N/A |
| 5341805 | 12/1993 | Stavridi et al.   | N/A | N/A |
| D353195 | 12/1993 | Savage et al.     | N/A | N/A |
| D353196 | 12/1993 | Savage et al.     | N/A | N/A |
| 5377676 | 12/1994 | Vari et al.       | N/A | N/A |
| 5387122 | 12/1994 | Goldberger et al. | N/A | N/A |
| 5397247 | 12/1994 | Aoki et al.       | N/A | N/A |
| D359546 | 12/1994 | Savage et al.     | N/A | N/A |
| 5431170 | 12/1994 | Mathews           | N/A | N/A |
| 5436499 | 12/1994 | Namavar et al.    | N/A | N/A |
| D361840 | 12/1994 | Savage et al.     | N/A | N/A |
| 5437275 | 12/1994 | Amundsen et al.   | N/A | N/A |
| D362063 | 12/1994 | Savage et al.     | N/A | N/A |
| 5452717 | 12/1994 | Branigan et al.   | N/A | N/A |
| D363120 | 12/1994 | Savage et al.     | N/A | N/A |
| 5456252 | 12/1994 | Vari et al.       | N/A | N/A |
| 5460182 | 12/1994 | Goodman et al.    | N/A | N/A |
| 5479934 | 12/1995 | Imran             | N/A | N/A |
| 5482036 | 12/1995 | Diab et al.       | N/A | N/A |
| 5490505 | 12/1995 | Diab et al.       | N/A | N/A |
| 5494043 | 12/1995 | O'Sullivan et al. | N/A | N/A |
| 5507286 | 12/1995 | Solenberger       | N/A | N/A |
| 5520177 | 12/1995 | Ogawa et al.      | N/A | N/A |
| 5533511 | 12/1995 | Kaspari et al.    | N/A | N/A |
| 5534851 | 12/1995 | Russek            | N/A | N/A |
| 5561275 | 12/1995 | Savage et al.     | N/A | N/A |

|         |         |                          |     |     |
|---------|---------|--------------------------|-----|-----|
| 5562002 | 12/1995 | Lalin                    | N/A | N/A |
| 5579373 | 12/1995 | Jang                     | N/A | N/A |
| 5590649 | 12/1996 | Caro et al.              | N/A | N/A |
| 5602924 | 12/1996 | Durand et al.            | N/A | N/A |
| 5619992 | 12/1996 | Guthrie et al.           | N/A | N/A |
| 5632272 | 12/1996 | Diab et al.              | N/A | N/A |
| 5638816 | 12/1996 | Kiani-Azarbayjany et al. | N/A | N/A |
| 5638818 | 12/1996 | Diab et al.              | N/A | N/A |
| 5645440 | 12/1996 | Tobler et al.            | N/A | N/A |
| 5660567 | 12/1996 | Nierlich et al.          | N/A | N/A |
| 5664270 | 12/1996 | Bell et al.              | N/A | N/A |
| 5671914 | 12/1996 | Kalkhoran et al.         | N/A | N/A |
| 5673693 | 12/1996 | Solenberger              | N/A | N/A |
| 5678544 | 12/1996 | DeLonzor et al.          | N/A | N/A |
| 5685299 | 12/1996 | Diab et al.              | N/A | N/A |
| 5726440 | 12/1997 | Kalkhoran et al.         | N/A | N/A |
| D393830 | 12/1997 | Tobler et al.            | N/A | N/A |
| 5740019 | 12/1997 | Lee                      | N/A | N/A |
| 5743262 | 12/1997 | Lepper, Jr. et al.       | N/A | N/A |
| 5747806 | 12/1997 | Khalil et al.            | N/A | N/A |
| 5750994 | 12/1997 | Schlager                 | N/A | N/A |
| 5758644 | 12/1997 | Diab et al.              | N/A | N/A |
| 5760910 | 12/1997 | Lepper, Jr. et al.       | N/A | N/A |
| 5769785 | 12/1997 | Diab et al.              | N/A | N/A |
| 5782757 | 12/1997 | Diab et al.              | N/A | N/A |
| 5785659 | 12/1997 | Caro et al.              | N/A | N/A |
| 5786592 | 12/1997 | Hök                      | N/A | N/A |
| 5791347 | 12/1997 | Flaherty et al.          | N/A | N/A |
| 5810724 | 12/1997 | Gronvall                 | N/A | N/A |
| 5810734 | 12/1997 | Caro et al.              | N/A | N/A |
| 5817008 | 12/1997 | Rafert et al.            | N/A | N/A |
| 5817010 | 12/1997 | Hibl                     | N/A | N/A |
| 5823950 | 12/1997 | Diab et al.              | N/A | N/A |
| 5830131 | 12/1997 | Caro et al.              | N/A | N/A |
| 5833618 | 12/1997 | Caro et al.              | N/A | N/A |
| RE36000 | 12/1997 | Swedlow et al.           | N/A | N/A |
| 5860919 | 12/1998 | Kiani-Azarbayjany et al. | N/A | N/A |
| 5879373 | 12/1998 | Röper et al.             | N/A | N/A |
| 5890929 | 12/1998 | Mills et al.             | N/A | N/A |
| 5904654 | 12/1998 | Wohltmann et al.         | N/A | N/A |
| 5910108 | 12/1998 | Solenberger              | N/A | N/A |
| 5919133 | 12/1998 | Taylor et al.            | N/A | N/A |
| 5919134 | 12/1998 | Diab                     | N/A | N/A |
| 5934925 | 12/1998 | Tobler et al.            | N/A | N/A |
| 5940182 | 12/1998 | Lepper, Jr. et al.       | N/A | N/A |
| 5987343 | 12/1998 | Kinast                   | N/A | N/A |
| 5991648 | 12/1998 | Levin                    | N/A | N/A |
| 5995855 | 12/1998 | Kiani et al.             | N/A | N/A |

|         |         |                          |     |     |
|---------|---------|--------------------------|-----|-----|
| 5997343 | 12/1998 | Mills et al.             | N/A | N/A |
| 5999834 | 12/1998 | Wang et al.              | N/A | N/A |
| 6002952 | 12/1998 | Diab et al.              | N/A | N/A |
| 6010937 | 12/1999 | Karam et al.             | N/A | N/A |
| 6011986 | 12/1999 | Diab et al.              | N/A | N/A |
| 6014576 | 12/1999 | Raley                    | N/A | N/A |
| 6027452 | 12/1999 | Flaherty et al.          | N/A | N/A |
| 6036642 | 12/1999 | Diab et al.              | N/A | N/A |
| 6040578 | 12/1999 | Malin et al.             | N/A | N/A |
| 6045509 | 12/1999 | Caro et al.              | N/A | N/A |
| 6061584 | 12/1999 | Lovejoy et al.           | N/A | N/A |
| 6066204 | 12/1999 | Haven                    | N/A | N/A |
| 6067462 | 12/1999 | Diab et al.              | N/A | N/A |
| 6081735 | 12/1999 | Diab et al.              | N/A | N/A |
| 6088607 | 12/1999 | Diab et al.              | N/A | N/A |
| 6110522 | 12/1999 | Lepper, Jr. et al.       | N/A | N/A |
| 6115673 | 12/1999 | Malin et al.             | N/A | N/A |
| 6124597 | 12/1999 | Shehada et al.           | N/A | N/A |
| 6128521 | 12/1999 | Marro et al.             | N/A | N/A |
| 6129675 | 12/1999 | Jay                      | N/A | N/A |
| 6144868 | 12/1999 | Parker                   | N/A | N/A |
| 6151516 | 12/1999 | Kiani-Azarbayjany et al. | N/A | N/A |
| 6152754 | 12/1999 | Gerhardt et al.          | N/A | N/A |
| 6157850 | 12/1999 | Diab et al.              | N/A | N/A |
| 6165005 | 12/1999 | Mills et al.             | N/A | N/A |
| 6184521 | 12/2000 | Coffin, IV et al.        | N/A | N/A |
| 6206830 | 12/2000 | Diab et al.              | N/A | N/A |
| 6229856 | 12/2000 | Diab et al.              | N/A | N/A |
| 6232609 | 12/2000 | Snyder et al.            | N/A | N/A |
| 6236872 | 12/2000 | Diab et al.              | N/A | N/A |
| 6241683 | 12/2000 | Macklem et al.           | N/A | N/A |
| 6253097 | 12/2000 | Aronow et al.            | N/A | N/A |
| 6255708 | 12/2000 | Sudharsanan et al.       | N/A | N/A |
| 6256523 | 12/2000 | Diab et al.              | N/A | N/A |
| 6263222 | 12/2000 | Diab et al.              | N/A | N/A |
| 6278522 | 12/2000 | Lepper, Jr. et al.       | N/A | N/A |
| 6280213 | 12/2000 | Tobler et al.            | N/A | N/A |
| 6280381 | 12/2000 | Malin et al.             | N/A | N/A |
| 6285896 | 12/2000 | Tobler et al.            | N/A | N/A |
| 6301493 | 12/2000 | Marro et al.             | N/A | N/A |
| 6308089 | 12/2000 | von der Ruhr et al.      | N/A | N/A |
| 6317627 | 12/2000 | Ennen et al.             | N/A | N/A |
| 6321000 | 12/2000 | King                     | N/A | N/A |
| 6321100 | 12/2000 | Parker                   | N/A | N/A |
| 6325761 | 12/2000 | Jay                      | N/A | N/A |
| 6334065 | 12/2000 | Al-Ali et al.            | N/A | N/A |
| 6343224 | 12/2001 | Parker                   | N/A | N/A |
| 6349228 | 12/2001 | Kiani et al.             | N/A | N/A |
| 6360114 | 12/2001 | Diab et al.              | N/A | N/A |

|         |         |                    |     |     |
|---------|---------|--------------------|-----|-----|
| 6368283 | 12/2001 | Xu et al.          | N/A | N/A |
| 6371921 | 12/2001 | Caro et al.        | N/A | N/A |
| 6377829 | 12/2001 | Al-Ali             | N/A | N/A |
| 6381489 | 12/2001 | Ashibe             | N/A | N/A |
| 6388240 | 12/2001 | Schulz et al.      | N/A | N/A |
| 6397091 | 12/2001 | Diab et al.        | N/A | N/A |
| 6411373 | 12/2001 | Garside et al.     | N/A | N/A |
| 6415167 | 12/2001 | Blank et al.       | N/A | N/A |
| 6430437 | 12/2001 | Marro              | N/A | N/A |
| 6430525 | 12/2001 | Weber et al.       | N/A | N/A |
| 6463311 | 12/2001 | Diab               | N/A | N/A |
| 6470199 | 12/2001 | Kopotic et al.     | N/A | N/A |
| 6487429 | 12/2001 | Hockersmith et al. | N/A | N/A |
| 6501975 | 12/2001 | Diab et al.        | N/A | N/A |
| 6505059 | 12/2002 | Kollias et al.     | N/A | N/A |
| 6515273 | 12/2002 | Al-Ali             | N/A | N/A |
| 6519484 | 12/2002 | Lovejoy et al.     | N/A | N/A |
| 6519487 | 12/2002 | Parker             | N/A | N/A |
| 6525386 | 12/2002 | Mills et al.       | N/A | N/A |
| 6526300 | 12/2002 | Kiani et al.       | N/A | N/A |
| 6534012 | 12/2002 | Hazen et al.       | N/A | N/A |
| 6541756 | 12/2002 | Schulz et al.      | N/A | N/A |
| 6542764 | 12/2002 | Al-Ali et al.      | N/A | N/A |
| 6571113 | 12/2002 | Fein et al.        | N/A | N/A |
| 6580086 | 12/2002 | Schulz et al.      | N/A | N/A |
| 6580948 | 12/2002 | Hauptert et al.    | N/A | N/A |
| 6584336 | 12/2002 | Ali et al.         | N/A | N/A |
| 6587196 | 12/2002 | Stippick et al.    | N/A | N/A |
| 6587199 | 12/2002 | Luu                | N/A | N/A |
| 6595316 | 12/2002 | Cybulski et al.    | N/A | N/A |
| 6597932 | 12/2002 | Tian et al.        | N/A | N/A |
| 6597933 | 12/2002 | Kiani et al.       | N/A | N/A |
| 6600940 | 12/2002 | Fein et al.        | N/A | N/A |
| 6606511 | 12/2002 | Ali et al.         | N/A | N/A |
| 6632181 | 12/2002 | Flaherty et al.    | N/A | N/A |
| 6635559 | 12/2002 | Greenwald et al.   | N/A | N/A |
| 6639668 | 12/2002 | Trepagnier         | N/A | N/A |
| 6640116 | 12/2002 | Diab               | N/A | N/A |
| 6640117 | 12/2002 | Makarewicz et al.  | N/A | N/A |
| 6643530 | 12/2002 | Diab et al.        | N/A | N/A |
| 6650917 | 12/2002 | Diab et al.        | N/A | N/A |
| 6654621 | 12/2002 | Palatnik           | N/A | N/A |
| 6654624 | 12/2002 | Diab et al.        | N/A | N/A |
| 6658276 | 12/2002 | Kiani et al.       | N/A | N/A |
| 6661161 | 12/2002 | Lanzo et al.       | N/A | N/A |
| 6671531 | 12/2002 | Al-Ali             | N/A | N/A |
| 6671532 | 12/2002 | Fudge et al.       | N/A | N/A |
| 6678543 | 12/2003 | Diab et al.        | N/A | N/A |
| 6684090 | 12/2003 | Ali et al.         | N/A | N/A |
| 6684091 | 12/2003 | Parker             | N/A | N/A |

|         |         |                          |     |     |
|---------|---------|--------------------------|-----|-----|
| 6697656 | 12/2003 | Al-Ali                   | N/A | N/A |
| 6697657 | 12/2003 | Shehada et al.           | N/A | N/A |
| 6697658 | 12/2003 | Al-Ali                   | N/A | N/A |
| RE38476 | 12/2003 | Diab et al.              | N/A | N/A |
| 6699194 | 12/2003 | Diab et al.              | N/A | N/A |
| 6714804 | 12/2003 | Al-Ali et al.            | N/A | N/A |
| RE38492 | 12/2003 | Diab et al.              | N/A | N/A |
| 6721582 | 12/2003 | Trepagnier et al.        | N/A | N/A |
| 6721585 | 12/2003 | Parker                   | N/A | N/A |
| 6725075 | 12/2003 | Al-Ali                   | N/A | N/A |
| 6728560 | 12/2003 | Kollias et al.           | N/A | N/A |
| 6735459 | 12/2003 | Parker                   | N/A | N/A |
| 6738652 | 12/2003 | Mattu et al.             | N/A | N/A |
| 6745060 | 12/2003 | Diab et al.              | N/A | N/A |
| 6760607 | 12/2003 | Al-Ali                   | N/A | N/A |
| 6770028 | 12/2003 | Ali et al.               | N/A | N/A |
| 6771994 | 12/2003 | Kiani et al.             | N/A | N/A |
| 6788965 | 12/2003 | Ruchti et al.            | N/A | N/A |
| 6792300 | 12/2003 | Diab et al.              | N/A | N/A |
| 6813511 | 12/2003 | Diab et al.              | N/A | N/A |
| 6816241 | 12/2003 | Grubisic                 | N/A | N/A |
| 6816741 | 12/2003 | Diab                     | N/A | N/A |
| 6822564 | 12/2003 | Al-Ali                   | N/A | N/A |
| 6826419 | 12/2003 | Diab et al.              | N/A | N/A |
| 6830711 | 12/2003 | Mills et al.             | N/A | N/A |
| 6839583 | 12/2004 | Lewandowski et al.       | N/A | N/A |
| 6850787 | 12/2004 | Weber et al.             | N/A | N/A |
| 6850788 | 12/2004 | Al-Ali                   | N/A | N/A |
| 6852083 | 12/2004 | Caro et al.              | N/A | N/A |
| 6861639 | 12/2004 | Al-Ali                   | N/A | N/A |
| 6876931 | 12/2004 | Lorenz et al.            | N/A | N/A |
| 6898452 | 12/2004 | Al-Ali et al.            | N/A | N/A |
| 6920345 | 12/2004 | Al-Ali et al.            | N/A | N/A |
| 6931268 | 12/2004 | Kiani-Azarbayjany et al. | N/A | N/A |
| 6934570 | 12/2004 | Kiani et al.             | N/A | N/A |
| 6939305 | 12/2004 | Flaherty et al.          | N/A | N/A |
| 6941162 | 12/2004 | Fudge et al.             | N/A | N/A |
| 6943348 | 12/2004 | Coffin IV                | N/A | N/A |
| 6950687 | 12/2004 | Al-Ali                   | N/A | N/A |
| 6956649 | 12/2004 | Acosta et al.            | N/A | N/A |
| 6961598 | 12/2004 | Diab                     | N/A | N/A |
| 6970792 | 12/2004 | Diab                     | N/A | N/A |
| 6979812 | 12/2004 | Al-Ali                   | N/A | N/A |
| 6985764 | 12/2005 | Mason et al.             | N/A | N/A |
| 6990364 | 12/2005 | Ruchti et al.            | N/A | N/A |
| 6993371 | 12/2005 | Kiani et al.             | N/A | N/A |
| 6996427 | 12/2005 | Ali et al.               | N/A | N/A |
| 6998247 | 12/2005 | Monfre et al.            | N/A | N/A |
| 6999904 | 12/2005 | Weber et al.             | N/A | N/A |

|         |         |                          |     |     |
|---------|---------|--------------------------|-----|-----|
| 7003338 | 12/2003 | Weber et al.             | N/A | N/A |
| 7003339 | 12/2005 | Diab et al.              | N/A | N/A |
| 7015451 | 12/2005 | Dalke et al.             | N/A | N/A |
| 7024233 | 12/2005 | Ali et al.               | N/A | N/A |
| 7027849 | 12/2005 | Al-Ali                   | N/A | N/A |
| 7030749 | 12/2005 | Al-Ali                   | N/A | N/A |
| 7039449 | 12/2005 | Al-Ali                   | N/A | N/A |
| 7041060 | 12/2005 | Flaherty et al.          | N/A | N/A |
| 7044918 | 12/2005 | Diab                     | N/A | N/A |
| 7048687 | 12/2005 | Reuss et al.             | N/A | N/A |
| 7067893 | 12/2005 | Mills et al.             | N/A | N/A |
| D526719 | 12/2005 | Richie, Jr. et al.       | N/A | N/A |
| 7096052 | 12/2005 | Mason et al.             | N/A | N/A |
| 7096054 | 12/2005 | Abdul-Hafiz et al.       | N/A | N/A |
| D529616 | 12/2005 | Deros et al.             | N/A | N/A |
| 7132641 | 12/2005 | Schulz et al.            | N/A | N/A |
| 7133710 | 12/2005 | Acosta et al.            | N/A | N/A |
| 7142901 | 12/2005 | Kiani et al.             | N/A | N/A |
| 7149561 | 12/2005 | Diab                     | N/A | N/A |
| 7186966 | 12/2006 | Al-Ali                   | N/A | N/A |
| 7190261 | 12/2006 | Al-Ali                   | N/A | N/A |
| 7215984 | 12/2006 | Diab et al.              | N/A | N/A |
| 7215986 | 12/2006 | Diab et al.              | N/A | N/A |
| 7221971 | 12/2006 | Diab et al.              | N/A | N/A |
| 7225006 | 12/2006 | Al-Ali et al.            | N/A | N/A |
| 7225007 | 12/2006 | Al-Ali et al.            | N/A | N/A |
| RE39672 | 12/2006 | Shehada et al.           | N/A | N/A |
| 7239905 | 12/2006 | Kiani-Azarbayjany et al. | N/A | N/A |
| 7245953 | 12/2006 | Parker                   | N/A | N/A |
| 7254429 | 12/2006 | Schurman et al.          | N/A | N/A |
| 7254431 | 12/2006 | Al-Ali et al.            | N/A | N/A |
| 7254433 | 12/2006 | Diab et al.              | N/A | N/A |
| 7254434 | 12/2006 | Schulz et al.            | N/A | N/A |
| 7272425 | 12/2006 | Al-Ali                   | N/A | N/A |
| 7274955 | 12/2006 | Kiani et al.             | N/A | N/A |
| D554263 | 12/2006 | Al-Ali et al.            | N/A | N/A |
| 7280858 | 12/2006 | Al-Ali et al.            | N/A | N/A |
| 7289835 | 12/2006 | Mansfield et al.         | N/A | N/A |
| 7292883 | 12/2006 | De Felice et al.         | N/A | N/A |
| 7295866 | 12/2006 | Al-Ali                   | N/A | N/A |
| 7328053 | 12/2007 | Diab et al.              | N/A | N/A |
| 7332784 | 12/2007 | Mills et al.             | N/A | N/A |
| 7340287 | 12/2007 | Mason et al.             | N/A | N/A |
| 7341559 | 12/2007 | Schulz et al.            | N/A | N/A |
| 7343186 | 12/2007 | Lamego et al.            | N/A | N/A |
| D566282 | 12/2007 | Al-Ali et al.            | N/A | N/A |
| 7355512 | 12/2007 | Al-Ali                   | N/A | N/A |
| 7356365 | 12/2007 | Schurman                 | N/A | N/A |
| 7371981 | 12/2007 | Abdul-Hafiz              | N/A | N/A |



|         |         |                 |     |     |
|---------|---------|-----------------|-----|-----|
| 7373193 | 12/2007 | Al-Ali et al.   | N/A | N/A |
| 7373194 | 12/2007 | Weber et al.    | N/A | N/A |
| 7376453 | 12/2007 | Diab et al.     | N/A | N/A |
| 7377794 | 12/2007 | Al-Ali et al.   | N/A | N/A |
| 7377899 | 12/2007 | Weber et al.    | N/A | N/A |
| 7383070 | 12/2007 | Diab et al.     | N/A | N/A |
| 7395158 | 12/2007 | Monfre et al.   | N/A | N/A |
| 7415297 | 12/2007 | Al-Ali et al.   | N/A | N/A |
| 7428432 | 12/2007 | Ali et al.      | N/A | N/A |
| 7438683 | 12/2007 | Al-Ali et al.   | N/A | N/A |
| 7440787 | 12/2007 | Diab            | N/A | N/A |
| 7454240 | 12/2007 | Diab et al.     | N/A | N/A |
| 7467002 | 12/2007 | Weber et al.    | N/A | N/A |
| 7469157 | 12/2007 | Diab et al.     | N/A | N/A |
| 7471969 | 12/2007 | Diab et al.     | N/A | N/A |
| 7471971 | 12/2007 | Diab et al.     | N/A | N/A |
| 7483729 | 12/2008 | Al-Ali et al.   | N/A | N/A |
| 7483730 | 12/2008 | Diab et al.     | N/A | N/A |
| 7489958 | 12/2008 | Diab et al.     | N/A | N/A |
| 7496391 | 12/2008 | Diab et al.     | N/A | N/A |
| 7496393 | 12/2008 | Diab et al.     | N/A | N/A |
| D587657 | 12/2008 | Al-Ali et al.   | N/A | N/A |
| 7499739 | 12/2008 | Sweitzer        | N/A | N/A |
| 7499741 | 12/2008 | Diab et al.     | N/A | N/A |
| 7499835 | 12/2008 | Weber et al.    | N/A | N/A |
| 7500950 | 12/2008 | Al-Ali et al.   | N/A | N/A |
| 7509154 | 12/2008 | Diab et al.     | N/A | N/A |
| 7509494 | 12/2008 | Al-Ali          | N/A | N/A |
| 7510849 | 12/2008 | Schurman et al. | N/A | N/A |
| 7514725 | 12/2008 | Wojtczuk et al. | N/A | N/A |
| 7519406 | 12/2008 | Blank et al.    | N/A | N/A |
| 7526328 | 12/2008 | Diab et al.     | N/A | N/A |
| D592507 | 12/2008 | Wachman et al.  | N/A | N/A |
| 7530942 | 12/2008 | Diab            | N/A | N/A |
| 7530949 | 12/2008 | Al Ali et al.   | N/A | N/A |
| 7530955 | 12/2008 | Diab et al.     | N/A | N/A |
| 7563110 | 12/2008 | Al-Ali et al.   | N/A | N/A |
| 7593230 | 12/2008 | Abul-Haj et al. | N/A | N/A |
| 7596398 | 12/2008 | Al-Ali et al.   | N/A | N/A |
| 7606608 | 12/2008 | Blank et al.    | N/A | N/A |
| 7618375 | 12/2008 | Flaherty et al. | N/A | N/A |
| 7620674 | 12/2008 | Ruchti et al.   | N/A | N/A |
| D606659 | 12/2008 | Kiani et al.    | N/A | N/A |
| 7629039 | 12/2008 | Eckerbom et al. | N/A | N/A |
| 7640140 | 12/2008 | Ruchti et al.   | N/A | N/A |
| 7647083 | 12/2009 | Al-Ali et al.   | N/A | N/A |
| D609193 | 12/2009 | Al-Ali et al.   | N/A | N/A |
| D614305 | 12/2009 | Al-Ali et al.   | N/A | N/A |
| 7697966 | 12/2009 | Monfre et al.   | N/A | N/A |
| 7698105 | 12/2009 | Ruchti et al.   | N/A | N/A |

|         |         |                          |     |     |
|---------|---------|--------------------------|-----|-----|
| 7706853 | 12/2009 | Hacker et al.            | N/A | N/A |
| RE41317 | 12/2009 | Parker                   | N/A | N/A |
| RE41333 | 12/2009 | Blank et al.             | N/A | N/A |
| 7729733 | 12/2009 | Al-Ali et al.            | N/A | N/A |
| 7734320 | 12/2009 | Al-Ali                   | N/A | N/A |
| 7761127 | 12/2009 | Al-Ali et al.            | N/A | N/A |
| 7761128 | 12/2009 | Al-Ali et al.            | N/A | N/A |
| 7764982 | 12/2009 | Dalke et al.             | N/A | N/A |
| D621516 | 12/2009 | Kiani et al.             | N/A | N/A |
| 7791155 | 12/2009 | Diab                     | N/A | N/A |
| 7801581 | 12/2009 | Diab                     | N/A | N/A |
| 7822452 | 12/2009 | Schurman et al.          | N/A | N/A |
| RE41912 | 12/2009 | Parker                   | N/A | N/A |
| 7844313 | 12/2009 | Kiani et al.             | N/A | N/A |
| 7844314 | 12/2009 | Al-Ali                   | N/A | N/A |
| 7844315 | 12/2009 | Al-Ali                   | N/A | N/A |
| 7865222 | 12/2010 | Weber et al.             | N/A | N/A |
| 7873497 | 12/2010 | Weber et al.             | N/A | N/A |
| 7880606 | 12/2010 | Al-Ali                   | N/A | N/A |
| 7880626 | 12/2010 | Al-Ali et al.            | N/A | N/A |
| 7891355 | 12/2010 | Al-Ali et al.            | N/A | N/A |
| 7894868 | 12/2010 | Al-Ali et al.            | N/A | N/A |
| 7899507 | 12/2010 | Al-Ali et al.            | N/A | N/A |
| 7899518 | 12/2010 | Trepagnier et al.        | N/A | N/A |
| 7904132 | 12/2010 | Weber et al.             | N/A | N/A |
| 7909772 | 12/2010 | Popov et al.             | N/A | N/A |
| 7910875 | 12/2010 | Al-Ali                   | N/A | N/A |
| 7919713 | 12/2010 | Al-Ali et al.            | N/A | N/A |
| 7937128 | 12/2010 | Al-Ali                   | N/A | N/A |
| 7937129 | 12/2010 | Mason et al.             | N/A | N/A |
| 7937130 | 12/2010 | Diab et al.              | N/A | N/A |
| 7941199 | 12/2010 | Kiani                    | N/A | N/A |
| 7951086 | 12/2010 | Flaherty et al.          | N/A | N/A |
| 7957780 | 12/2010 | Lamego et al.            | N/A | N/A |
| 7962188 | 12/2010 | Kiani et al.             | N/A | N/A |
| 7962190 | 12/2010 | Diab et al.              | N/A | N/A |
| 7976472 | 12/2010 | Kiani                    | N/A | N/A |
| 7988637 | 12/2010 | Diab                     | N/A | N/A |
| 7990382 | 12/2010 | Kiani                    | N/A | N/A |
| 7991446 | 12/2010 | Ali et al.               | N/A | N/A |
| 8000761 | 12/2010 | Al-Ali                   | N/A | N/A |
| 8008088 | 12/2010 | Bellott et al.           | N/A | N/A |
| RE42753 | 12/2010 | Kiani-Azarbayjany et al. | N/A | N/A |
| 8019400 | 12/2010 | Diab et al.              | N/A | N/A |
| 8028701 | 12/2010 | Al-Ali et al.            | N/A | N/A |
| 8029765 | 12/2010 | Bellott et al.           | N/A | N/A |
| 8036727 | 12/2010 | Schurman et al.          | N/A | N/A |
| 8036728 | 12/2010 | Diab et al.              | N/A | N/A |
| 8046040 | 12/2010 | Ali et al.               | N/A | N/A |

|         |         |                      |     |     |
|---------|---------|----------------------|-----|-----|
| 8046041 | 12/2010 | Diab et al.          | N/A | N/A |
| 8046042 | 12/2010 | Diab et al.          | N/A | N/A |
| 8048040 | 12/2010 | Kiani                | N/A | N/A |
| 8050728 | 12/2010 | Al-Ali et al.        | N/A | N/A |
| RE43169 | 12/2011 | Parker               | N/A | N/A |
| 8118620 | 12/2011 | Al-Ali et al.        | N/A | N/A |
| 8126528 | 12/2011 | Diab et al.          | N/A | N/A |
| 8128572 | 12/2011 | Diab et al.          | N/A | N/A |
| 8130105 | 12/2011 | Al-Ali et al.        | N/A | N/A |
| 8145287 | 12/2011 | Diab et al.          | N/A | N/A |
| 8150487 | 12/2011 | Diab et al.          | N/A | N/A |
| 8175672 | 12/2011 | Parker               | N/A | N/A |
| 8180420 | 12/2011 | Diab et al.          | N/A | N/A |
| 8182443 | 12/2011 | Kiani                | N/A | N/A |
| 8185180 | 12/2011 | Diab et al.          | N/A | N/A |
| 8190223 | 12/2011 | Al-Ali et al.        | N/A | N/A |
| 8190227 | 12/2011 | Diab et al.          | N/A | N/A |
| 8203438 | 12/2011 | Kiani et al.         | N/A | N/A |
| 8203704 | 12/2011 | Merritt et al.       | N/A | N/A |
| 8204566 | 12/2011 | Schurman et al.      | N/A | N/A |
| 8219172 | 12/2011 | Schurman et al.      | N/A | N/A |
| 8224411 | 12/2011 | Al-Ali et al.        | N/A | N/A |
| 8228181 | 12/2011 | Al-Ali               | N/A | N/A |
| 8229532 | 12/2011 | Davis                | N/A | N/A |
| 8229533 | 12/2011 | Diab et al.          | N/A | N/A |
| 8233955 | 12/2011 | Al-Ali et al.        | N/A | N/A |
| 8244325 | 12/2011 | Al-Ali et al.        | N/A | N/A |
| 8255026 | 12/2011 | Al-Ali               | N/A | N/A |
| 8255027 | 12/2011 | Al-Ali et al.        | N/A | N/A |
| 8255028 | 12/2011 | Al-Ali et al.        | N/A | N/A |
| 8260577 | 12/2011 | Weber et al.         | N/A | N/A |
| 8265723 | 12/2011 | McHale et al.        | N/A | N/A |
| 8274360 | 12/2011 | Sampath et al.       | N/A | N/A |
| 8280473 | 12/2011 | Al-Ali               | N/A | N/A |
| 8301217 | 12/2011 | Al-Ali et al.        | N/A | N/A |
| 8306596 | 12/2011 | Schurman et al.      | N/A | N/A |
| 8310336 | 12/2011 | Muhsin et al.        | N/A | N/A |
| 8315683 | 12/2011 | Al-Ali et al.        | N/A | N/A |
| RE43860 | 12/2011 | Parker               | N/A | N/A |
| 8337403 | 12/2011 | Al-Ali et al.        | N/A | N/A |
| 8346330 | 12/2012 | Lamego               | N/A | N/A |
| 8353842 | 12/2012 | Al-Ali et al.        | N/A | N/A |
| 8355766 | 12/2012 | MacNeish, III et al. | N/A | N/A |
| 8359080 | 12/2012 | Diab et al.          | N/A | N/A |
| 8364223 | 12/2012 | Al-Ali et al.        | N/A | N/A |
| 8364226 | 12/2012 | Diab et al.          | N/A | N/A |
| 8374665 | 12/2012 | Lamego               | N/A | N/A |
| 8385995 | 12/2012 | Al-Ali et al.        | N/A | N/A |
| 8385996 | 12/2012 | Smith et al.         | N/A | N/A |
| 8388353 | 12/2012 | Kiani et al.         | N/A | N/A |

|         |         |                    |     |     |
|---------|---------|--------------------|-----|-----|
| 8399822 | 12/2012 | Al-Ali             | N/A | N/A |
| 8401602 | 12/2012 | Kiani              | N/A | N/A |
| 8405608 | 12/2012 | Al-Ali et al.      | N/A | N/A |
| 8414499 | 12/2012 | Al-Ali et al.      | N/A | N/A |
| 8418524 | 12/2012 | Al-Ali             | N/A | N/A |
| 8423106 | 12/2012 | Lamego et al.      | N/A | N/A |
| 8428967 | 12/2012 | Olsen et al.       | N/A | N/A |
| 8430817 | 12/2012 | Al-Ali et al.      | N/A | N/A |
| 8437825 | 12/2012 | Dalvi et al.       | N/A | N/A |
| 8455290 | 12/2012 | Siskavich          | N/A | N/A |
| 8457703 | 12/2012 | Al-Ali             | N/A | N/A |
| 8457707 | 12/2012 | Kiani              | N/A | N/A |
| 8463349 | 12/2012 | Diab et al.        | N/A | N/A |
| 8466286 | 12/2012 | Bellott et al.     | N/A | N/A |
| 8471713 | 12/2012 | Poeze et al.       | N/A | N/A |
| 8473020 | 12/2012 | Kiani et al.       | N/A | N/A |
| 8483787 | 12/2012 | Al-Ali et al.      | N/A | N/A |
| 8489364 | 12/2012 | Weber et al.       | N/A | N/A |
| 8498684 | 12/2012 | Weber et al.       | N/A | N/A |
| 8504128 | 12/2012 | Blank et al.       | N/A | N/A |
| 8509867 | 12/2012 | Workman et al.     | N/A | N/A |
| 8515509 | 12/2012 | Bruinsma et al.    | N/A | N/A |
| 8523781 | 12/2012 | Al-Ali             | N/A | N/A |
| 8529301 | 12/2012 | Al-Ali et al.      | N/A | N/A |
| 8532727 | 12/2012 | Ali et al.         | N/A | N/A |
| 8532728 | 12/2012 | Diab et al.        | N/A | N/A |
| D692145 | 12/2012 | Al-Ali et al.      | N/A | N/A |
| 8547209 | 12/2012 | Kiani et al.       | N/A | N/A |
| 8548548 | 12/2012 | Al-Ali             | N/A | N/A |
| 8548549 | 12/2012 | Schurman et al.    | N/A | N/A |
| 8548550 | 12/2012 | Al-Ali et al.      | N/A | N/A |
| 8560032 | 12/2012 | Al-Ali et al.      | N/A | N/A |
| 8560034 | 12/2012 | Diab et al.        | N/A | N/A |
| 8570167 | 12/2012 | Al-Ali             | N/A | N/A |
| 8570503 | 12/2012 | Vo et al.          | N/A | N/A |
| 8571617 | 12/2012 | Reichgott et al.   | N/A | N/A |
| 8571618 | 12/2012 | Lamego et al.      | N/A | N/A |
| 8571619 | 12/2012 | Al-Ali et al.      | N/A | N/A |
| 8577431 | 12/2012 | Lamego et al.      | N/A | N/A |
| 8581732 | 12/2012 | Al-Ali et al.      | N/A | N/A |
| 8584345 | 12/2012 | Al-Ali et al.      | N/A | N/A |
| 8588880 | 12/2012 | Abdul-Hafiz et al. | N/A | N/A |
| 8600467 | 12/2012 | Al-Ali et al.      | N/A | N/A |
| 8606342 | 12/2012 | Diab               | N/A | N/A |
| 8626255 | 12/2013 | Al-Ali et al.      | N/A | N/A |
| 8630691 | 12/2013 | Lamego et al.      | N/A | N/A |
| 8634889 | 12/2013 | Al-Ali et al.      | N/A | N/A |
| 8641631 | 12/2013 | Sierra et al.      | N/A | N/A |
| 8652060 | 12/2013 | Al-Ali             | N/A | N/A |
| 8663107 | 12/2013 | Kiani              | N/A | N/A |

|         |         |                      |     |     |
|---------|---------|----------------------|-----|-----|
| 8666468 | 12/2013 | Al-Ali               | N/A | N/A |
| 8667967 | 12/2013 | Al-Ali et al.        | N/A | N/A |
| 8670811 | 12/2013 | O'Reilly             | N/A | N/A |
| 8670814 | 12/2013 | Diab et al.          | N/A | N/A |
| 8676286 | 12/2013 | Weber et al.         | N/A | N/A |
| 8682407 | 12/2013 | Al-Ali               | N/A | N/A |
| RE44823 | 12/2013 | Parker               | N/A | N/A |
| RE44875 | 12/2013 | Kiani et al.         | N/A | N/A |
| 8688183 | 12/2013 | Bruinsma et al.      | N/A | N/A |
| 8690799 | 12/2013 | Telfort et al.       | N/A | N/A |
| 8700112 | 12/2013 | Kiani                | N/A | N/A |
| 8702627 | 12/2013 | Telfort et al.       | N/A | N/A |
| 8706179 | 12/2013 | Parker               | N/A | N/A |
| 8712494 | 12/2013 | MacNeish, III et al. | N/A | N/A |
| 8715206 | 12/2013 | Telfort et al.       | N/A | N/A |
| 8718735 | 12/2013 | Lamego et al.        | N/A | N/A |
| 8718737 | 12/2013 | Diab et al.          | N/A | N/A |
| 8718738 | 12/2013 | Blank et al.         | N/A | N/A |
| 8720249 | 12/2013 | Al-Ali               | N/A | N/A |
| 8721541 | 12/2013 | Al-Ali et al.        | N/A | N/A |
| 8721542 | 12/2013 | Al-Ali et al.        | N/A | N/A |
| 8723677 | 12/2013 | Kiani                | N/A | N/A |
| 8740792 | 12/2013 | Kiani et al.         | N/A | N/A |
| 8754776 | 12/2013 | Poeze et al.         | N/A | N/A |
| 8755535 | 12/2013 | Telfort et al.       | N/A | N/A |
| 8755856 | 12/2013 | Diab et al.          | N/A | N/A |
| 8755872 | 12/2013 | Marinow              | N/A | N/A |
| 8761850 | 12/2013 | Lamego               | N/A | N/A |
| 8764671 | 12/2013 | Kiani                | N/A | N/A |
| 8768423 | 12/2013 | Shakespeare et al.   | N/A | N/A |
| 8771204 | 12/2013 | Telfort et al.       | N/A | N/A |
| 8777634 | 12/2013 | Kiani et al.         | N/A | N/A |
| 8781543 | 12/2013 | Diab et al.          | N/A | N/A |
| 8781544 | 12/2013 | Al-Ali et al.        | N/A | N/A |
| 8781549 | 12/2013 | Al-Ali et al.        | N/A | N/A |
| 8788003 | 12/2013 | Schurman et al.      | N/A | N/A |
| 8790268 | 12/2013 | Al-Ali               | N/A | N/A |
| 8801613 | 12/2013 | Al-Ali et al.        | N/A | N/A |
| 8821397 | 12/2013 | Al-Ali et al.        | N/A | N/A |
| 8821415 | 12/2013 | Al-Ali et al.        | N/A | N/A |
| 8830449 | 12/2013 | Lamego et al.        | N/A | N/A |
| 8831700 | 12/2013 | Schurman et al.      | N/A | N/A |
| 8840549 | 12/2013 | Al-Ali et al.        | N/A | N/A |
| 8847740 | 12/2013 | Kiani et al.         | N/A | N/A |
| 8849365 | 12/2013 | Smith et al.         | N/A | N/A |
| 8852094 | 12/2013 | Al-Ali et al.        | N/A | N/A |
| 8852994 | 12/2013 | Wojtczuk et al.      | N/A | N/A |
| 8868147 | 12/2013 | Stippick et al.      | N/A | N/A |
| 8868150 | 12/2013 | Al-Ali et al.        | N/A | N/A |
| 8870792 | 12/2013 | Al-Ali et al.        | N/A | N/A |

|         |         |                  |     |     |
|---------|---------|------------------|-----|-----|
| 8886271 | 12/2013 | Kiani et al.     | N/A | N/A |
| 8888539 | 12/2013 | Al-Ali et al.    | N/A | N/A |
| 8888708 | 12/2013 | Diab et al.      | N/A | N/A |
| 8892180 | 12/2013 | Weber et al.     | N/A | N/A |
| 8897847 | 12/2013 | Al-Ali           | N/A | N/A |
| 8909310 | 12/2013 | Lamego et al.    | N/A | N/A |
| 8911377 | 12/2013 | Al-Ali           | N/A | N/A |
| 8912909 | 12/2013 | Al-Ali et al.    | N/A | N/A |
| 8920317 | 12/2013 | Al-Ali et al.    | N/A | N/A |
| 8921699 | 12/2013 | Al-Ali et al.    | N/A | N/A |
| 8922382 | 12/2013 | Al-Ali et al.    | N/A | N/A |
| 8929964 | 12/2014 | Al-Ali et al.    | N/A | N/A |
| 8942777 | 12/2014 | Diab et al.      | N/A | N/A |
| 8948834 | 12/2014 | Diab et al.      | N/A | N/A |
| 8948835 | 12/2014 | Diab             | N/A | N/A |
| 8965471 | 12/2014 | Lamego           | N/A | N/A |
| 8983564 | 12/2014 | Al-Ali           | N/A | N/A |
| 8989831 | 12/2014 | Al-Ali et al.    | N/A | N/A |
| 8996085 | 12/2014 | Kiani et al.     | N/A | N/A |
| 8998809 | 12/2014 | Kiani            | N/A | N/A |
| 9028429 | 12/2014 | Telfort et al.   | N/A | N/A |
| 9037207 | 12/2014 | Al-Ali et al.    | N/A | N/A |
| 9060721 | 12/2014 | Reichgott et al. | N/A | N/A |
| 9066666 | 12/2014 | Kiani            | N/A | N/A |
| 9066680 | 12/2014 | Al-Ali et al.    | N/A | N/A |
| 9072474 | 12/2014 | Al-Ali et al.    | N/A | N/A |
| 9078560 | 12/2014 | Schurman et al.  | N/A | N/A |
| 9084569 | 12/2014 | Weber et al.     | N/A | N/A |
| 9095316 | 12/2014 | Welch et al.     | N/A | N/A |
| 9106038 | 12/2014 | Telfort et al.   | N/A | N/A |
| 9107625 | 12/2014 | Telfort et al.   | N/A | N/A |
| 9107626 | 12/2014 | Al-Ali et al.    | N/A | N/A |
| 9113831 | 12/2014 | Al-Ali           | N/A | N/A |
| 9113832 | 12/2014 | Al-Ali           | N/A | N/A |
| 9119595 | 12/2014 | Lamego           | N/A | N/A |
| 9131881 | 12/2014 | Diab et al.      | N/A | N/A |
| 9131882 | 12/2014 | Al-Ali et al.    | N/A | N/A |
| 9131883 | 12/2014 | Al-Ali           | N/A | N/A |
| 9131917 | 12/2014 | Telfort et al.   | N/A | N/A |
| 9138180 | 12/2014 | Coverston et al. | N/A | N/A |
| 9138182 | 12/2014 | Al-Ali et al.    | N/A | N/A |
| 9138192 | 12/2014 | Weber et al.     | N/A | N/A |
| 9142117 | 12/2014 | Muhsin et al.    | N/A | N/A |
| 9153112 | 12/2014 | Kiani et al.     | N/A | N/A |
| 9153121 | 12/2014 | Kiani et al.     | N/A | N/A |
| 9161696 | 12/2014 | Al-Ali et al.    | N/A | N/A |
| 9161713 | 12/2014 | Al-Ali et al.    | N/A | N/A |
| 9167995 | 12/2014 | Lamego et al.    | N/A | N/A |
| 9176141 | 12/2014 | Al-Ali et al.    | N/A | N/A |
| 9186102 | 12/2014 | Bruinsma et al.  | N/A | N/A |

|         |         |                    |     |     |
|---------|---------|--------------------|-----|-----|
| 9192312 | 12/2014 | Al-Ali             | N/A | N/A |
| 9192329 | 12/2014 | Al-Ali             | N/A | N/A |
| 9192351 | 12/2014 | Telfort et al.     | N/A | N/A |
| 9195385 | 12/2014 | Al-Ali et al.      | N/A | N/A |
| 9211072 | 12/2014 | Kiani              | N/A | N/A |
| 9211095 | 12/2014 | Al-Ali             | N/A | N/A |
| 9218454 | 12/2014 | Kiani et al.       | N/A | N/A |
| 9226696 | 12/2015 | Kiani              | N/A | N/A |
| 9241662 | 12/2015 | Al-Ali et al.      | N/A | N/A |
| 9245668 | 12/2015 | Vo et al.          | N/A | N/A |
| 9259185 | 12/2015 | Abdul-Hafiz et al. | N/A | N/A |
| 9267572 | 12/2015 | Barker et al.      | N/A | N/A |
| 9277880 | 12/2015 | Poeze et al.       | N/A | N/A |
| 9289167 | 12/2015 | Diab et al.        | N/A | N/A |
| 9295421 | 12/2015 | Kiani et al.       | N/A | N/A |
| 9307928 | 12/2015 | Al-Ali et al.      | N/A | N/A |
| 9323894 | 12/2015 | Kiani              | N/A | N/A |
| D755392 | 12/2015 | Hwang et al.       | N/A | N/A |
| 9326712 | 12/2015 | Kiani              | N/A | N/A |
| 9333316 | 12/2015 | Kiani              | N/A | N/A |
| 9339220 | 12/2015 | Lamego et al.      | N/A | N/A |
| 9341565 | 12/2015 | Lamego et al.      | N/A | N/A |
| 9351673 | 12/2015 | Diab et al.        | N/A | N/A |
| 9351675 | 12/2015 | Al-Ali et al.      | N/A | N/A |
| 9364181 | 12/2015 | Kiani et al.       | N/A | N/A |
| 9368671 | 12/2015 | Wojtczuk et al.    | N/A | N/A |
| 9370325 | 12/2015 | Al-Ali et al.      | N/A | N/A |
| 9370326 | 12/2015 | McHale et al.      | N/A | N/A |
| 9370335 | 12/2015 | Al-Ali et al.      | N/A | N/A |
| 9375185 | 12/2015 | Ali et al.         | N/A | N/A |
| 9386953 | 12/2015 | Al-Ali             | N/A | N/A |
| 9386961 | 12/2015 | Al-Ali et al.      | N/A | N/A |
| 9392945 | 12/2015 | Al-Ali et al.      | N/A | N/A |
| 9397448 | 12/2015 | Al-Ali et al.      | N/A | N/A |
| 9408542 | 12/2015 | Kinast et al.      | N/A | N/A |
| 9436645 | 12/2015 | Al-Ali et al.      | N/A | N/A |
| 9445759 | 12/2015 | Lamego et al.      | N/A | N/A |
| 9466919 | 12/2015 | Kiani et al.       | N/A | N/A |
| 9474474 | 12/2015 | Lamego et al.      | N/A | N/A |
| 9480422 | 12/2015 | Al-Ali             | N/A | N/A |
| 9480435 | 12/2015 | Olsen              | N/A | N/A |
| 9492110 | 12/2015 | Al-Ali et al.      | N/A | N/A |
| 9510779 | 12/2015 | Poeze et al.       | N/A | N/A |
| 9517024 | 12/2015 | Kiani et al.       | N/A | N/A |
| 9532722 | 12/2016 | Lamego et al.      | N/A | N/A |
| 9538949 | 12/2016 | Al-Ali et al.      | N/A | N/A |
| 9538980 | 12/2016 | Telfort et al.     | N/A | N/A |
| 9549696 | 12/2016 | Lamego et al.      | N/A | N/A |
| 9554737 | 12/2016 | Schurman et al.    | N/A | N/A |
| 9560996 | 12/2016 | Kiani              | N/A | N/A |

|         |         |                    |     |     |
|---------|---------|--------------------|-----|-----|
| 9560998 | 12/2016 | Al-Ali et al.      | N/A | N/A |
| 9566019 | 12/2016 | Al-Ali et al.      | N/A | N/A |
| 9579039 | 12/2016 | Jansen et al.      | N/A | N/A |
| 9591975 | 12/2016 | Dalvi et al.       | N/A | N/A |
| 9622692 | 12/2016 | Lamego et al.      | N/A | N/A |
| 9622693 | 12/2016 | Diab               | N/A | N/A |
| D788312 | 12/2016 | Al-Ali et al.      | N/A | N/A |
| 9636055 | 12/2016 | Al Ali et al.      | N/A | N/A |
| 9636056 | 12/2016 | Al-Ali             | N/A | N/A |
| 9649054 | 12/2016 | Lamego et al.      | N/A | N/A |
| 9662052 | 12/2016 | Al-Ali et al.      | N/A | N/A |
| 9668679 | 12/2016 | Schurman et al.    | N/A | N/A |
| 9668680 | 12/2016 | Bruinsma et al.    | N/A | N/A |
| 9668703 | 12/2016 | Al-Ali             | N/A | N/A |
| 9675286 | 12/2016 | Diab               | N/A | N/A |
| 9687160 | 12/2016 | Kiani              | N/A | N/A |
| 9693719 | 12/2016 | Al-Ali et al.      | N/A | N/A |
| 9693737 | 12/2016 | Al-Ali             | N/A | N/A |
| 9697928 | 12/2016 | Al-Ali et al.      | N/A | N/A |
| 9717425 | 12/2016 | Kiani et al.       | N/A | N/A |
| 9717458 | 12/2016 | Lamego et al.      | N/A | N/A |
| 9724016 | 12/2016 | Al-Ali et al.      | N/A | N/A |
| 9724024 | 12/2016 | Al-Ali             | N/A | N/A |
| 9724025 | 12/2016 | Kiani et al.       | N/A | N/A |
| 9730640 | 12/2016 | Diab et al.        | N/A | N/A |
| 9743887 | 12/2016 | Al-Ali et al.      | N/A | N/A |
| 9749232 | 12/2016 | Sampath et al.     | N/A | N/A |
| 9750442 | 12/2016 | Olsen              | N/A | N/A |
| 9750443 | 12/2016 | Smith et al.       | N/A | N/A |
| 9750461 | 12/2016 | Telfort            | N/A | N/A |
| 9775545 | 12/2016 | Al-Ali et al.      | N/A | N/A |
| 9775546 | 12/2016 | Diab et al.        | N/A | N/A |
| 9775570 | 12/2016 | Al-Ali             | N/A | N/A |
| 9778079 | 12/2016 | Al-Ali et al.      | N/A | N/A |
| 9782077 | 12/2016 | Lamego et al.      | N/A | N/A |
| 9782110 | 12/2016 | Kiani              | N/A | N/A |
| 9787568 | 12/2016 | Lamego et al.      | N/A | N/A |
| 9788735 | 12/2016 | Al-Ali             | N/A | N/A |
| 9788768 | 12/2016 | Al-Ali et al.      | N/A | N/A |
| 9795300 | 12/2016 | Al-Ali             | N/A | N/A |
| 9795310 | 12/2016 | Al-Ali             | N/A | N/A |
| 9795358 | 12/2016 | Telfort et al.     | N/A | N/A |
| 9795739 | 12/2016 | Al-Ali et al.      | N/A | N/A |
| 9801556 | 12/2016 | Kiani              | N/A | N/A |
| 9801588 | 12/2016 | Weber et al.       | N/A | N/A |
| 9808188 | 12/2016 | Perea et al.       | N/A | N/A |
| 9814418 | 12/2016 | Weber et al.       | N/A | N/A |
| 9820691 | 12/2016 | Kiani              | N/A | N/A |
| 9833152 | 12/2016 | Kiani et al.       | N/A | N/A |
| 9833180 | 12/2016 | Shakespeare et al. | N/A | N/A |



|          |         |                  |     |     |
|----------|---------|------------------|-----|-----|
| 9839379  | 12/2016 | Al-Ali et al.    | N/A | N/A |
| 9839381  | 12/2016 | Weber et al.     | N/A | N/A |
| 9847002  | 12/2016 | Kiani et al.     | N/A | N/A |
| 9847749  | 12/2016 | Kiani et al.     | N/A | N/A |
| 9848800  | 12/2016 | Lee et al.       | N/A | N/A |
| 9848806  | 12/2016 | Al-Ali           | N/A | N/A |
| 9848807  | 12/2016 | Lamego           | N/A | N/A |
| 9861298  | 12/2017 | Eckerbom et al.  | N/A | N/A |
| 9861304  | 12/2017 | Al-Ali et al.    | N/A | N/A |
| 9861305  | 12/2017 | Weber et al.     | N/A | N/A |
| 9867578  | 12/2017 | Al-Ali et al.    | N/A | N/A |
| 9872623  | 12/2017 | Al-Ali           | N/A | N/A |
| 9876320  | 12/2017 | Coverston et al. | N/A | N/A |
| 9877650  | 12/2017 | Muhsin et al.    | N/A | N/A |
| 9877686  | 12/2017 | Al-Ali et al.    | N/A | N/A |
| 9891079  | 12/2017 | Dalvi            | N/A | N/A |
| 9895107  | 12/2017 | Al-Ali et al.    | N/A | N/A |
| 9913617  | 12/2017 | Al-Ali et al.    | N/A | N/A |
| 9924893  | 12/2017 | Schurman et al.  | N/A | N/A |
| 9924897  | 12/2017 | Abdul-Hafiz      | N/A | N/A |
| 9936917  | 12/2017 | Poeze et al.     | N/A | N/A |
| 9943269  | 12/2017 | Muhsin et al.    | N/A | N/A |
| 9949676  | 12/2017 | Al-Ali           | N/A | N/A |
| 9955937  | 12/2017 | Telfort          | N/A | N/A |
| 9965946  | 12/2017 | Al-Ali et al.    | N/A | N/A |
| 9980667  | 12/2017 | Kiani et al.     | N/A | N/A |
| D820865  | 12/2017 | Muhsin et al.    | N/A | N/A |
| 9986919  | 12/2017 | Lamego et al.    | N/A | N/A |
| 9986952  | 12/2017 | Dalvi et al.     | N/A | N/A |
| 9989560  | 12/2017 | Poeze et al.     | N/A | N/A |
| 9993207  | 12/2017 | Al-Ali et al.    | N/A | N/A |
| 10007758 | 12/2017 | Al-Ali et al.    | N/A | N/A |
| D822215  | 12/2017 | Al-Ali et al.    | N/A | N/A |
| D822216  | 12/2017 | Barker et al.    | N/A | N/A |
| 10010276 | 12/2017 | Al-Ali et al.    | N/A | N/A |
| 10032002 | 12/2017 | Kiani et al.     | N/A | N/A |
| 10039482 | 12/2017 | Al-Ali et al.    | N/A | N/A |
| 10052037 | 12/2017 | Kinast et al.    | N/A | N/A |
| 10058275 | 12/2017 | Al-Ali et al.    | N/A | N/A |
| 10064562 | 12/2017 | Al-Ali           | N/A | N/A |
| 10086138 | 12/2017 | Novak, Jr.       | N/A | N/A |
| 10092200 | 12/2017 | Al-Ali et al.    | N/A | N/A |
| 10092249 | 12/2017 | Kiani et al.     | N/A | N/A |
| 10098550 | 12/2017 | Al-Ali et al.    | N/A | N/A |
| 10098591 | 12/2017 | Al-Ali et al.    | N/A | N/A |
| 10098610 | 12/2017 | Al-Ali et al.    | N/A | N/A |
| 10111591 | 12/2017 | Dyell et al.     | N/A | N/A |
| D833624  | 12/2017 | DeJong et al.    | N/A | N/A |
| 10123726 | 12/2017 | Al-Ali et al.    | N/A | N/A |
| 10123729 | 12/2017 | Dyell et al.     | N/A | N/A |

|          |         |                 |     |     |
|----------|---------|-----------------|-----|-----|
| 10130289 | 12/2017 | Al-Ali et al.   | N/A | N/A |
| 10130291 | 12/2017 | Schurman et al. | N/A | N/A |
| D835282  | 12/2017 | Barker et al.   | N/A | N/A |
| D835283  | 12/2017 | Barker et al.   | N/A | N/A |
| D835284  | 12/2017 | Barker et al.   | N/A | N/A |
| D835285  | 12/2017 | Barker et al.   | N/A | N/A |
| 10149616 | 12/2017 | Al-Ali et al.   | N/A | N/A |
| 10154815 | 12/2017 | Al-Ali et al.   | N/A | N/A |
| 10159412 | 12/2017 | Lamego et al.   | N/A | N/A |
| 10188296 | 12/2018 | Al-Ali et al.   | N/A | N/A |
| 10188331 | 12/2018 | Kiani et al.    | N/A | N/A |
| 10188348 | 12/2018 | Al-Ali et al.   | N/A | N/A |
| RE47218  | 12/2018 | Al-Ali          | N/A | N/A |
| RE47244  | 12/2018 | Kiani et al.    | N/A | N/A |
| RE47249  | 12/2018 | Kiani et al.    | N/A | N/A |
| 10194847 | 12/2018 | Al-Ali          | N/A | N/A |
| 10194848 | 12/2018 | Kiani et al.    | N/A | N/A |
| 10201298 | 12/2018 | Al-Ali et al.   | N/A | N/A |
| 10205272 | 12/2018 | Kiani et al.    | N/A | N/A |
| 10205291 | 12/2018 | Scruggs et al.  | N/A | N/A |
| 10213108 | 12/2018 | Al-Ali          | N/A | N/A |
| 10219706 | 12/2018 | Al-Ali          | N/A | N/A |
| 10219746 | 12/2018 | McHale et al.   | N/A | N/A |
| 10226187 | 12/2018 | Al-Ali et al.   | N/A | N/A |
| 10226576 | 12/2018 | Kiani           | N/A | N/A |
| 10231657 | 12/2018 | Al-Ali et al.   | N/A | N/A |
| 10231670 | 12/2018 | Blank et al.    | N/A | N/A |
| 10231676 | 12/2018 | Al-Ali et al.   | N/A | N/A |
| RE47353  | 12/2018 | Kiani et al.    | N/A | N/A |
| 10251585 | 12/2018 | Al-Ali et al.   | N/A | N/A |
| 10251586 | 12/2018 | Lamego          | N/A | N/A |
| 10255994 | 12/2018 | Sampath et al.  | N/A | N/A |
| 10258265 | 12/2018 | Poeze et al.    | N/A | N/A |
| 10258266 | 12/2018 | Poeze et al.    | N/A | N/A |
| 10271748 | 12/2018 | Al-Ali          | N/A | N/A |
| 10279247 | 12/2018 | Kiani           | N/A | N/A |
| 10292664 | 12/2018 | Al-Ali          | N/A | N/A |
| 10299720 | 12/2018 | Brown et al.    | N/A | N/A |
| 10327337 | 12/2018 | Schmidt et al.  | N/A | N/A |
| 10327713 | 12/2018 | Barker et al.   | N/A | N/A |
| 10332630 | 12/2018 | Al-Ali          | N/A | N/A |
| 10383520 | 12/2018 | Wojtczuk et al. | N/A | N/A |
| 10383527 | 12/2018 | Al-Ali          | N/A | N/A |
| 10388120 | 12/2018 | Muhsin et al.   | N/A | N/A |
| D864120  | 12/2018 | Forrest et al.  | N/A | N/A |
| 10441181 | 12/2018 | Telfort et al.  | N/A | N/A |
| 10441196 | 12/2018 | Eckerbom et al. | N/A | N/A |
| 10448844 | 12/2018 | Al-Ali et al.   | N/A | N/A |
| 10448871 | 12/2018 | Al-Ali et al.   | N/A | N/A |
| 10456038 | 12/2018 | Lamego et al.   | N/A | N/A |

|          |         |                    |     |     |
|----------|---------|--------------------|-----|-----|
| 10463340 | 12/2018 | Telfort et al.     | N/A | N/A |
| 10471159 | 12/2018 | Lapotko et al.     | N/A | N/A |
| 10505311 | 12/2018 | Al-Ali et al.      | N/A | N/A |
| 10524738 | 12/2019 | Olsen              | N/A | N/A |
| 10532174 | 12/2019 | Al-Ali             | N/A | N/A |
| 10537285 | 12/2019 | Shreim et al.      | N/A | N/A |
| 10542903 | 12/2019 | Al-Ali et al.      | N/A | N/A |
| 10555678 | 12/2019 | Dalvi et al.       | N/A | N/A |
| 10568553 | 12/2019 | O'Neil et al.      | N/A | N/A |
| RE47882  | 12/2019 | Al-Ali             | N/A | N/A |
| 10608817 | 12/2019 | Haider et al.      | N/A | N/A |
| D880477  | 12/2019 | Forrest et al.     | N/A | N/A |
| 10617302 | 12/2019 | Al-Ali et al.      | N/A | N/A |
| 10617335 | 12/2019 | Al-Ali et al.      | N/A | N/A |
| 10637181 | 12/2019 | Al-Ali et al.      | N/A | N/A |
| D886849  | 12/2019 | Muhsin et al.      | N/A | N/A |
| D887548  | 12/2019 | Abdul-Hafiz et al. | N/A | N/A |
| D887549  | 12/2019 | Abdul-Hafiz et al. | N/A | N/A |
| 10667764 | 12/2019 | Ahmed et al.       | N/A | N/A |
| D890708  | 12/2019 | Forrest et al.     | N/A | N/A |
| 10721785 | 12/2019 | Al-Ali             | N/A | N/A |
| 10736518 | 12/2019 | Al-Ali et al.      | N/A | N/A |
| 10750984 | 12/2019 | Pauley et al.      | N/A | N/A |
| D897098  | 12/2019 | Al-Ali             | N/A | N/A |
| 10779098 | 12/2019 | Iswanto et al.     | N/A | N/A |
| 10827961 | 12/2019 | Iyengar et al.     | N/A | N/A |
| 10828007 | 12/2019 | Telfort et al.     | N/A | N/A |
| 10832818 | 12/2019 | Muhsin et al.      | N/A | N/A |
| 10849554 | 12/2019 | Shreim et al.      | N/A | N/A |
| 10856750 | 12/2019 | Indorf             | N/A | N/A |
| D906970  | 12/2020 | Forrest et al.     | N/A | N/A |
| D908213  | 12/2020 | Abdul-Hafiz et al. | N/A | N/A |
| 10918281 | 12/2020 | Al-Ali et al.      | N/A | N/A |
| 10932705 | 12/2020 | Muhsin et al.      | N/A | N/A |
| 10932729 | 12/2020 | Kiani et al.       | N/A | N/A |
| 10939878 | 12/2020 | Kiani et al.       | N/A | N/A |
| 10956950 | 12/2020 | Al-Ali et al.      | N/A | N/A |
| D916135  | 12/2020 | Indorf et al.      | N/A | N/A |
| D917046  | 12/2020 | Abdul-Hafiz et al. | N/A | N/A |
| D917550  | 12/2020 | Indorf et al.      | N/A | N/A |
| D917564  | 12/2020 | Indorf et al.      | N/A | N/A |
| D917704  | 12/2020 | Al-Ali et al.      | N/A | N/A |
| 10987066 | 12/2020 | Chandran et al.    | N/A | N/A |
| 10991135 | 12/2020 | Al-Ali et al.      | N/A | N/A |
| D919094  | 12/2020 | Al-Ali et al.      | N/A | N/A |
| D919100  | 12/2020 | Al-Ali et al.      | N/A | N/A |
| 11006867 | 12/2020 | Al-Ali             | N/A | N/A |
| D921202  | 12/2020 | Al-Ali et al.      | N/A | N/A |
| 11024064 | 12/2020 | Muhsin et al.      | N/A | N/A |
| 11026604 | 12/2020 | Chen et al.        | N/A | N/A |

|          |         |                 |     |     |
|----------|---------|-----------------|-----|-----|
| D925597  | 12/2020 | Chandran et al. | N/A | N/A |
| D927699  | 12/2020 | Al-Ali et al.   | N/A | N/A |
| 11076777 | 12/2020 | Lee et al.      | N/A | N/A |
| 11114188 | 12/2020 | Poeze et al.    | N/A | N/A |
| D933232  | 12/2020 | Al-Ali et al.   | N/A | N/A |
| D933233  | 12/2020 | Al-Ali et al.   | N/A | N/A |
| D933234  | 12/2020 | Al-Ali et al.   | N/A | N/A |
| 11145408 | 12/2020 | Sampath et al.  | N/A | N/A |
| 11147518 | 12/2020 | Al-Ali et al.   | N/A | N/A |
| 11185262 | 12/2020 | Al-Ali et al.   | N/A | N/A |
| 11191484 | 12/2020 | Kiani et al.    | N/A | N/A |
| D946596  | 12/2021 | Ahmed           | N/A | N/A |
| D946597  | 12/2021 | Ahmed           | N/A | N/A |
| D946598  | 12/2021 | Ahmed           | N/A | N/A |
| D946617  | 12/2021 | Ahmed           | N/A | N/A |
| 11272839 | 12/2021 | Al-Ali et al.   | N/A | N/A |
| 11289199 | 12/2021 | Al-Ali          | N/A | N/A |
| RE49034  | 12/2021 | Al-Ali          | N/A | N/A |
| 11298021 | 12/2021 | Muhsin et al.   | N/A | N/A |
| D950580  | 12/2021 | Ahmed           | N/A | N/A |
| D950599  | 12/2021 | Ahmed           | N/A | N/A |
| D950738  | 12/2021 | Al-Ali et al.   | N/A | N/A |
| 11331042 | 12/2021 | Al-Ali et al.   | N/A | N/A |
| D957648  | 12/2021 | Al-Ali          | N/A | N/A |
| 11382567 | 12/2021 | O'Brien et al.  | N/A | N/A |
| 11389093 | 12/2021 | Triman et al.   | N/A | N/A |
| 11406286 | 12/2021 | Al-Ali et al.   | N/A | N/A |
| 11417426 | 12/2021 | Muhsin et al.   | N/A | N/A |
| 11439329 | 12/2021 | Lamego          | N/A | N/A |
| 11445948 | 12/2021 | Scruggs et al.  | N/A | N/A |
| D965789  | 12/2021 | Al-Ali et al.   | N/A | N/A |
| D967433  | 12/2021 | Al-Ali et al.   | N/A | N/A |
| 11464410 | 12/2021 | Muhsin          | N/A | N/A |
| 11504058 | 12/2021 | Sharma et al.   | N/A | N/A |
| 11504066 | 12/2021 | Dalvi et al.    | N/A | N/A |
| D971933  | 12/2021 | Ahmed           | N/A | N/A |
| D973072  | 12/2021 | Ahmed           | N/A | N/A |
| D973685  | 12/2021 | Ahmed           | N/A | N/A |
| D973686  | 12/2021 | Ahmed           | N/A | N/A |
| D974193  | 12/2022 | Forrest et al.  | N/A | N/A |
| D979516  | 12/2022 | Al-Ali et al.   | N/A | N/A |
| D980091  | 12/2022 | Forrest et al.  | N/A | N/A |
| 11596363 | 12/2022 | Lamego          | N/A | N/A |
| 11627919 | 12/2022 | Kiani et al.    | N/A | N/A |
| 11637437 | 12/2022 | Al-Ali et al.   | N/A | N/A |
| D985498  | 12/2022 | Al-Ali et al.   | N/A | N/A |
| 11653862 | 12/2022 | Dalvi et al.    | N/A | N/A |
| D989112  | 12/2022 | Muhsin et al.   | N/A | N/A |
| D989327  | 12/2022 | Al-Ali et al.   | N/A | N/A |
| 11678829 | 12/2022 | Al-Ali et al.   | N/A | N/A |

|          |         |                      |     |     |
|----------|---------|----------------------|-----|-----|
| 11679579 | 12/2022 | Al-Ali               | N/A | N/A |
| 11684296 | 12/2022 | Vo et al.            | N/A | N/A |
| 11692934 | 12/2022 | Normand et al.       | N/A | N/A |
| 11701043 | 12/2022 | Al-Ali et al.        | N/A | N/A |
| D997365  | 12/2022 | Hwang                | N/A | N/A |
| 11721105 | 12/2022 | Ranasinghe et al.    | N/A | N/A |
| 11730379 | 12/2022 | Ahmed et al.         | N/A | N/A |
| D998625  | 12/2022 | Indorf et al.        | N/A | N/A |
| D998630  | 12/2022 | Indorf et al.        | N/A | N/A |
| D998631  | 12/2022 | Indorf et al.        | N/A | N/A |
| D999244  | 12/2022 | Indorf et al.        | N/A | N/A |
| D999245  | 12/2022 | Indorf et al.        | N/A | N/A |
| D999246  | 12/2022 | Indorf et al.        | N/A | N/A |
| 11766198 | 12/2022 | Pauley et al.        | N/A | N/A |
| D1000975 | 12/2022 | Al-Ali et al.        | N/A | N/A |
| 11803623 | 12/2022 | Kiani et al.         | N/A | N/A |
| 11832940 | 12/2022 | Diab et al.          | N/A | N/A |
| D1013179 | 12/2023 | Al-Ali et al.        | N/A | N/A |
| 11872156 | 12/2023 | Telfort et al.       | N/A | N/A |
| 11879960 | 12/2023 | Ranasinghe et al.    | N/A | N/A |
| 11883129 | 12/2023 | Olsen                | N/A | N/A |
| D1022729 | 12/2023 | Forrest et al.       | N/A | N/A |
| 11951186 | 12/2023 | Krishnamani et al.   | N/A | N/A |
| 11974833 | 12/2023 | Forrest et al.       | N/A | N/A |
| 11986067 | 12/2023 | Al-Ali et al.        | N/A | N/A |
| 11986289 | 12/2023 | Dalvi et al.         | N/A | N/A |
| 11986305 | 12/2023 | Al-Ali et al.        | N/A | N/A |
| D1031729 | 12/2023 | Forrest et al.       | N/A | N/A |
| 12004869 | 12/2023 | Kiani et al.         | N/A | N/A |
| 12014328 | 12/2023 | Wachman et al.       | N/A | N/A |
| D1036293 | 12/2023 | Al-Ali et al.        | N/A | N/A |
| D1037462 | 12/2023 | Al-Ali et al.        | N/A | N/A |
| 12029844 | 12/2023 | Pauley et al.        | N/A | N/A |
| 12048534 | 12/2023 | Vo et al.            | N/A | N/A |
| 12064217 | 12/2023 | Ahmed et al.         | N/A | N/A |
| 12066426 | 12/2023 | Lapotko et al.       | N/A | N/A |
| D1041511 | 12/2023 | Indorf et al.        | N/A | N/A |
| D1042596 | 12/2023 | DeJong et al.        | N/A | N/A |
| D1042852 | 12/2023 | Hwang                | N/A | N/A |
| 12076159 | 12/2023 | Belur Nagaraj et al. | N/A | N/A |
| 12082926 | 12/2023 | Sharma et al.        | N/A | N/A |
| D1044828 | 12/2023 | Chandran et al.      | N/A | N/A |
| D1048571 | 12/2023 | Yu et al.            | N/A | N/A |
| D1048908 | 12/2023 | Al-Ali et al.        | N/A | N/A |
| 12106752 | 12/2023 | Campbell et al.      | N/A | N/A |
| 12114974 | 12/2023 | Al-Ali et al.        | N/A | N/A |
| 12126683 | 12/2023 | Koo et al.           | N/A | N/A |
| 12127838 | 12/2023 | Olsen et al.         | N/A | N/A |
| 12128213 | 12/2023 | Kiani et al.         | N/A | N/A |
| 12131661 | 12/2023 | Pauley et al.        | N/A | N/A |

|              |         |                      |     |     |
|--------------|---------|----------------------|-----|-----|
| D1050910     | 12/2023 | Al-Ali et al.        | N/A | N/A |
| 2001/0034477 | 12/2000 | Mansfield et al.     | N/A | N/A |
| 2001/0039483 | 12/2000 | Brand et al.         | N/A | N/A |
| 2002/0010401 | 12/2001 | Bushmakina et al.    | N/A | N/A |
| 2002/0058864 | 12/2001 | Mansfield et al.     | N/A | N/A |
| 2002/0133080 | 12/2001 | Apruzzese et al.     | N/A | N/A |
| 2003/0013975 | 12/2002 | Kiani                | N/A | N/A |
| 2003/0018243 | 12/2002 | Gerhardt et al.      | N/A | N/A |
| 2003/0144582 | 12/2002 | Cohen et al.         | N/A | N/A |
| 2003/0156288 | 12/2002 | Barnum et al.        | N/A | N/A |
| 2003/0212312 | 12/2002 | Coffin, IV et al.    | N/A | N/A |
| 2004/0106163 | 12/2003 | Workman, Jr. et al.  | N/A | N/A |
| 2005/0055276 | 12/2004 | Kiani et al.         | N/A | N/A |
| 2005/0075550 | 12/2004 | Lindelkugel          | N/A | N/A |
| 2005/0234317 | 12/2004 | Kiani                | N/A | N/A |
| 2006/0015023 | 12/2005 | Monfre et al.        | N/A | N/A |
| 2006/0053522 | 12/2005 | Kimbell              | N/A | N/A |
| 2006/0073719 | 12/2005 | Kiani                | N/A | N/A |
| 2006/0161054 | 12/2005 | Reuss et al.         | N/A | N/A |
| 2006/0189871 | 12/2005 | Al-Ali et al.        | N/A | N/A |
| 2007/0073116 | 12/2006 | Kiani et al.         | N/A | N/A |
| 2007/0100219 | 12/2006 | Sweitzer             | N/A | N/A |
| 2007/0180140 | 12/2006 | Welch et al.         | N/A | N/A |
| 2007/0244377 | 12/2006 | Cozad et al.         | N/A | N/A |
| 2008/0064965 | 12/2007 | Jay et al.           | N/A | N/A |
| 2008/0076982 | 12/2007 | Ollerdersen et al.   | N/A | N/A |
| 2008/0088467 | 12/2007 | Al-Ali               | N/A | N/A |
| 2008/0094228 | 12/2007 | Welch et al.         | N/A | N/A |
| 2008/0103375 | 12/2007 | Kiani                | N/A | N/A |
| 2008/0221418 | 12/2007 | Al-Ali et al.        | N/A | N/A |
| 2008/0262324 | 12/2007 | Van Der Voort et al. | N/A | N/A |
| 2009/0036759 | 12/2008 | Ault et al.          | N/A | N/A |
| 2009/0093687 | 12/2008 | Telfort et al.       | N/A | N/A |
| 2009/0095926 | 12/2008 | MacNeish, III        | N/A | N/A |
| 2009/0247984 | 12/2008 | Lamego et al.        | N/A | N/A |
| 2009/0275813 | 12/2008 | Davis                | N/A | N/A |
| 2009/0275844 | 12/2008 | Al-Ali               | N/A | N/A |
| 2010/0004518 | 12/2009 | Vo et al.            | N/A | N/A |
| 2010/0030040 | 12/2009 | Poeze et al.         | N/A | N/A |
| 2010/0099964 | 12/2009 | O'Reilly et al.      | N/A | N/A |
| 2010/0234718 | 12/2009 | Sampath et al.       | N/A | N/A |
| 2010/0240972 | 12/2009 | Neal                 | N/A | N/A |
| 2010/0270257 | 12/2009 | Wachman et al.       | N/A | N/A |
| 2010/0305417 | 12/2009 | Kumazaki             | N/A | N/A |
| 2011/0028806 | 12/2010 | Merritt et al.       | N/A | N/A |
| 2011/0028809 | 12/2010 | Goodman              | N/A | N/A |
| 2011/0038754 | 12/2010 | James                | N/A | N/A |
| 2011/0040197 | 12/2010 | Welch et al.         | N/A | N/A |
| 2011/0082711 | 12/2010 | Poeze et al.         | N/A | N/A |
| 2011/0087081 | 12/2010 | Kiani et al.         | N/A | N/A |

|              |         |                    |     |     |
|--------------|---------|--------------------|-----|-----|
| 2011/0118561 | 12/2010 | Tari et al.        | N/A | N/A |
| 2011/0125060 | 12/2010 | Telfort et al.     | N/A | N/A |
| 2011/0137297 | 12/2010 | Kiani et al.       | N/A | N/A |
| 2011/0172498 | 12/2010 | Olsen et al.       | N/A | N/A |
| 2011/0208015 | 12/2010 | Welch et al.       | N/A | N/A |
| 2011/0230733 | 12/2010 | Al-Ali             | N/A | N/A |
| 2011/0261528 | 12/2010 | Gandhi             | N/A | N/A |
| 2012/0123231 | 12/2011 | O'Reilly           | N/A | N/A |
| 2012/0165629 | 12/2011 | Merritt et al.     | N/A | N/A |
| 2012/0202361 | 12/2011 | Terhune            | N/A | N/A |
| 2012/0209082 | 12/2011 | Al-Ali             | N/A | N/A |
| 2012/0209084 | 12/2011 | Olsen et al.       | N/A | N/A |
| 2012/0226117 | 12/2011 | Lamego et al.      | N/A | N/A |
| 2012/0283524 | 12/2011 | Kiani et al.       | N/A | N/A |
| 2013/0023775 | 12/2012 | Lamego et al.      | N/A | N/A |
| 2013/0041591 | 12/2012 | Lamego             | N/A | N/A |
| 2013/0060147 | 12/2012 | Welch et al.       | N/A | N/A |
| 2013/0096405 | 12/2012 | Garfio             | N/A | N/A |
| 2013/0096936 | 12/2012 | Sampath et al.     | N/A | N/A |
| 2013/0243021 | 12/2012 | Siskavich          | N/A | N/A |
| 2013/0296672 | 12/2012 | O'Neil et al.      | N/A | N/A |
| 2013/0324808 | 12/2012 | Al-Ali et al.      | N/A | N/A |
| 2013/0331660 | 12/2012 | Al-Ali et al.      | N/A | N/A |
| 2013/0345921 | 12/2012 | Al-Ali et al.      | N/A | N/A |
| 2014/0012100 | 12/2013 | Al-Ali et al.      | N/A | N/A |
| 2014/0051953 | 12/2013 | Lamego et al.      | N/A | N/A |
| 2014/0120564 | 12/2013 | Workman et al.     | N/A | N/A |
| 2014/0121482 | 12/2013 | Merritt et al.     | N/A | N/A |
| 2014/0127137 | 12/2013 | Bellott et al.     | N/A | N/A |
| 2014/0163344 | 12/2013 | Al-Ali             | N/A | N/A |
| 2014/0166076 | 12/2013 | Kiani et al.       | N/A | N/A |
| 2014/0171763 | 12/2013 | Diab               | N/A | N/A |
| 2014/0180154 | 12/2013 | Sierra et al.      | N/A | N/A |
| 2014/0180160 | 12/2013 | Brown et al.       | N/A | N/A |
| 2014/0187973 | 12/2013 | Brown et al.       | N/A | N/A |
| 2014/0213864 | 12/2013 | Abdul-Hafiz et al. | N/A | N/A |
| 2014/0275835 | 12/2013 | Lamego et al.      | N/A | N/A |
| 2014/0275871 | 12/2013 | Lamego et al.      | N/A | N/A |
| 2014/0275872 | 12/2013 | Merritt et al.     | N/A | N/A |
| 2014/0288400 | 12/2013 | Diab et al.        | N/A | N/A |
| 2014/0316217 | 12/2013 | Purdon et al.      | N/A | N/A |
| 2014/0316218 | 12/2013 | Purdon et al.      | N/A | N/A |
| 2014/0316228 | 12/2013 | Blank et al.       | N/A | N/A |
| 2014/0323825 | 12/2013 | Al-Ali et al.      | N/A | N/A |
| 2014/0323897 | 12/2013 | Brown et al.       | N/A | N/A |
| 2014/0323898 | 12/2013 | Purdon et al.      | N/A | N/A |
| 2014/0330098 | 12/2013 | Merritt et al.     | N/A | N/A |
| 2014/0357966 | 12/2013 | Al-Ali et al.      | N/A | N/A |
| 2015/0005600 | 12/2014 | Blank et al.       | N/A | N/A |
| 2015/0011907 | 12/2014 | Purdon et al.      | N/A | N/A |

|              |         |                  |     |     |
|--------------|---------|------------------|-----|-----|
| 2015/0032029 | 12/2014 | Al-Ali et al.    | N/A | N/A |
| 2015/0038859 | 12/2014 | Dalvi et al.     | N/A | N/A |
| 2015/0073241 | 12/2014 | Lamego           | N/A | N/A |
| 2015/0080754 | 12/2014 | Purdon et al.    | N/A | N/A |
| 2015/0087936 | 12/2014 | Al-Ali et al.    | N/A | N/A |
| 2015/0094546 | 12/2014 | Al-Ali           | N/A | N/A |
| 2015/0099950 | 12/2014 | Al-Ali et al.    | N/A | N/A |
| 2015/0101844 | 12/2014 | Al-Ali et al.    | N/A | N/A |
| 2015/0106121 | 12/2014 | Muhsin et al.    | N/A | N/A |
| 2015/0112151 | 12/2014 | Muhsin et al.    | N/A | N/A |
| 2015/0196249 | 12/2014 | Brown et al.     | N/A | N/A |
| 2015/0216459 | 12/2014 | Al-Ali et al.    | N/A | N/A |
| 2015/0238722 | 12/2014 | Al-Ali           | N/A | N/A |
| 2015/0257689 | 12/2014 | Al-Ali et al.    | N/A | N/A |
| 2015/0351697 | 12/2014 | Weber et al.     | N/A | N/A |
| 2015/0366507 | 12/2014 | Blank et al.     | N/A | N/A |
| 2016/0029932 | 12/2015 | Al-Ali           | N/A | N/A |
| 2016/0058347 | 12/2015 | Reichgott et al. | N/A | N/A |
| 2016/0066824 | 12/2015 | Al-Ali et al.    | N/A | N/A |
| 2016/0081552 | 12/2015 | Wojtczuk et al.  | N/A | N/A |
| 2016/0095543 | 12/2015 | Telfort et al.   | N/A | N/A |
| 2016/0103598 | 12/2015 | Al-Ali et al.    | N/A | N/A |
| 2016/0166183 | 12/2015 | Poeze et al.     | N/A | N/A |
| 2016/0196388 | 12/2015 | Lamego           | N/A | N/A |
| 2016/0197436 | 12/2015 | Barker et al.    | N/A | N/A |
| 2016/0213281 | 12/2015 | Eckerbom et al.  | N/A | N/A |
| 2016/0228043 | 12/2015 | O'Neil et al.    | N/A | N/A |
| 2016/0234944 | 12/2015 | Schmidt et al.   | N/A | N/A |
| 2016/0270735 | 12/2015 | Diab et al.      | N/A | N/A |
| 2016/0283665 | 12/2015 | Sampath et al.   | N/A | N/A |
| 2016/0287786 | 12/2015 | Kiani            | N/A | N/A |
| 2016/0314260 | 12/2015 | Kiani            | N/A | N/A |
| 2016/0324488 | 12/2015 | Olsen            | N/A | N/A |
| 2016/0327984 | 12/2015 | Al-Ali et al.    | N/A | N/A |
| 2016/0331332 | 12/2015 | Al-Ali           | N/A | N/A |
| 2016/0367173 | 12/2015 | Dalvi et al.     | N/A | N/A |
| 2017/0000394 | 12/2016 | Al-Ali et al.    | N/A | N/A |
| 2017/0007134 | 12/2016 | Al-Ali et al.    | N/A | N/A |
| 2017/0014083 | 12/2016 | Diab et al.      | N/A | N/A |
| 2017/0014084 | 12/2016 | Al-Ali et al.    | N/A | N/A |
| 2017/0024748 | 12/2016 | Haider           | N/A | N/A |
| 2017/0042488 | 12/2016 | Muhsin           | N/A | N/A |
| 2017/0055882 | 12/2016 | Al-Ali et al.    | N/A | N/A |
| 2017/0055887 | 12/2016 | Al-Ali           | N/A | N/A |
| 2017/0055896 | 12/2016 | Al-Ali           | N/A | N/A |
| 2017/0079594 | 12/2016 | Telfort et al.   | N/A | N/A |
| 2017/0086723 | 12/2016 | Al-Ali et al.    | N/A | N/A |
| 2017/0143281 | 12/2016 | Olsen            | N/A | N/A |
| 2017/0147774 | 12/2016 | Kiani            | N/A | N/A |
| 2017/0156620 | 12/2016 | Al-Ali et al.    | N/A | N/A |



|              |         |                    |     |     |
|--------------|---------|--------------------|-----|-----|
| 2017/0173632 | 12/2016 | Al-Ali             | N/A | N/A |
| 2017/0188919 | 12/2016 | Al-Ali et al.      | N/A | N/A |
| 2017/0196464 | 12/2016 | Jansen et al.      | N/A | N/A |
| 2017/0196470 | 12/2016 | Lamego et al.      | N/A | N/A |
| 2017/0228516 | 12/2016 | Sampath et al.     | N/A | N/A |
| 2017/0245790 | 12/2016 | Al-Ali et al.      | N/A | N/A |
| 2017/0251974 | 12/2016 | Shreim et al.      | N/A | N/A |
| 2017/0251975 | 12/2016 | Shreim et al.      | N/A | N/A |
| 2017/0258403 | 12/2016 | Abdul-Hafiz et al. | N/A | N/A |
| 2017/0311891 | 12/2016 | Kiani et al.       | N/A | N/A |
| 2017/0332976 | 12/2016 | Al-Ali             | N/A | N/A |
| 2017/0340293 | 12/2016 | Al-Ali et al.      | N/A | N/A |
| 2017/0360310 | 12/2016 | Kiani              | N/A | N/A |
| 2018/0008146 | 12/2017 | Al-Ali et al.      | N/A | N/A |
| 2018/0013562 | 12/2017 | Haider et al.      | N/A | N/A |
| 2018/0014752 | 12/2017 | Al-Ali et al.      | N/A | N/A |
| 2018/0028124 | 12/2017 | Al-Ali et al.      | N/A | N/A |
| 2018/0055390 | 12/2017 | Kiani et al.       | N/A | N/A |
| 2018/0055430 | 12/2017 | Diab et al.        | N/A | N/A |
| 2018/0064381 | 12/2017 | Shakespeare et al. | N/A | N/A |
| 2018/0069776 | 12/2017 | Lamego et al.      | N/A | N/A |
| 2018/0070867 | 12/2017 | Smith et al.       | N/A | N/A |
| 2018/0082767 | 12/2017 | Al-Ali et al.      | N/A | N/A |
| 2018/0085068 | 12/2017 | Telfort            | N/A | N/A |
| 2018/0087937 | 12/2017 | Al-Ali et al.      | N/A | N/A |
| 2018/0103874 | 12/2017 | Lee et al.         | N/A | N/A |
| 2018/0103905 | 12/2017 | Kiani              | N/A | N/A |
| 2018/0110478 | 12/2017 | Al-Ali             | N/A | N/A |
| 2018/0116575 | 12/2017 | Perea et al.       | N/A | N/A |
| 2018/0125368 | 12/2017 | Lamego et al.      | N/A | N/A |
| 2018/0125430 | 12/2017 | Al-Ali et al.      | N/A | N/A |
| 2018/0125445 | 12/2017 | Telfort et al.     | N/A | N/A |
| 2018/0130325 | 12/2017 | Kiani et al.       | N/A | N/A |
| 2018/0132769 | 12/2017 | Weber et al.       | N/A | N/A |
| 2018/0146901 | 12/2017 | Al-Ali et al.      | N/A | N/A |
| 2018/0146902 | 12/2017 | Kiani et al.       | N/A | N/A |
| 2018/0153442 | 12/2017 | Eckerbom et al.    | N/A | N/A |
| 2018/0153446 | 12/2017 | Kiani              | N/A | N/A |
| 2018/0153447 | 12/2017 | Al-Ali et al.      | N/A | N/A |
| 2018/0153448 | 12/2017 | Weber et al.       | N/A | N/A |
| 2018/0161499 | 12/2017 | Al-Ali et al.      | N/A | N/A |
| 2018/0168491 | 12/2017 | Al-Ali et al.      | N/A | N/A |
| 2018/0174679 | 12/2017 | Sampath et al.     | N/A | N/A |
| 2018/0174680 | 12/2017 | Sampath et al.     | N/A | N/A |
| 2018/0184917 | 12/2017 | Kiani              | N/A | N/A |
| 2018/0192924 | 12/2017 | Al-Ali             | N/A | N/A |
| 2018/0192953 | 12/2017 | Shreim et al.      | N/A | N/A |
| 2018/0192955 | 12/2017 | Al-Ali et al.      | N/A | N/A |
| 2018/0199871 | 12/2017 | Pauley et al.      | N/A | N/A |
| 2018/0206795 | 12/2017 | Al-Ali             | N/A | N/A |

|              |         |                 |     |     |
|--------------|---------|-----------------|-----|-----|
| 2018/0206815 | 12/2017 | Telfort         | N/A | N/A |
| 2018/0213583 | 12/2017 | Al-Ali          | N/A | N/A |
| 2018/0214031 | 12/2017 | Kiani et al.    | N/A | N/A |
| 2018/0214090 | 12/2017 | Al-Ali et al.   | N/A | N/A |
| 2018/0218792 | 12/2017 | Muhsin et al.   | N/A | N/A |
| 2018/0225960 | 12/2017 | Al-Ali et al.   | N/A | N/A |
| 2018/0238718 | 12/2017 | Dalvi           | N/A | N/A |
| 2018/0242853 | 12/2017 | Al-Ali          | N/A | N/A |
| 2018/0242921 | 12/2017 | Muhsin et al.   | N/A | N/A |
| 2018/0242923 | 12/2017 | Al-Ali et al.   | N/A | N/A |
| 2018/0242924 | 12/2017 | Barker et al.   | N/A | N/A |
| 2018/0242926 | 12/2017 | Muhsin et al.   | N/A | N/A |
| 2018/0247353 | 12/2017 | Al-Ali et al.   | N/A | N/A |
| 2018/0247712 | 12/2017 | Muhsin et al.   | N/A | N/A |
| 2018/0249933 | 12/2017 | Schurman et al. | N/A | N/A |
| 2018/0253947 | 12/2017 | Muhsin et al.   | N/A | N/A |
| 2018/0256087 | 12/2017 | Al-Ali et al.   | N/A | N/A |
| 2018/0256113 | 12/2017 | Weber et al.    | N/A | N/A |
| 2018/0285094 | 12/2017 | Housel et al.   | N/A | N/A |
| 2018/0289325 | 12/2017 | Poeze et al.    | N/A | N/A |
| 2018/0289337 | 12/2017 | Al-Ali et al.   | N/A | N/A |
| 2018/0296161 | 12/2017 | Shreim et al.   | N/A | N/A |
| 2018/0300919 | 12/2017 | Muhsin et al.   | N/A | N/A |
| 2018/0310822 | 12/2017 | Indorf et al.   | N/A | N/A |
| 2018/0310823 | 12/2017 | Al-Ali et al.   | N/A | N/A |
| 2018/0317826 | 12/2017 | Muhsin et al.   | N/A | N/A |
| 2018/0317841 | 12/2017 | Novak, Jr.      | N/A | N/A |
| 2018/0333055 | 12/2017 | Lamego et al.   | N/A | N/A |
| 2018/0333087 | 12/2017 | Al-Ali          | N/A | N/A |
| 2019/0000317 | 12/2018 | Muhsin et al.   | N/A | N/A |
| 2019/0000362 | 12/2018 | Kiani et al.    | N/A | N/A |
| 2019/0015023 | 12/2018 | Monfre          | N/A | N/A |
| 2019/0021638 | 12/2018 | Al-Ali et al.   | N/A | N/A |
| 2019/0029574 | 12/2018 | Schurman et al. | N/A | N/A |
| 2019/0029578 | 12/2018 | Al-Ali et al.   | N/A | N/A |
| 2019/0058280 | 12/2018 | Al-Ali et al.   | N/A | N/A |
| 2019/0058281 | 12/2018 | Al-Ali et al.   | N/A | N/A |
| 2019/0069813 | 12/2018 | Al-Ali          | N/A | N/A |
| 2019/0069814 | 12/2018 | Al-Ali          | N/A | N/A |
| 2019/0076028 | 12/2018 | Al-Ali et al.   | N/A | N/A |
| 2019/0082979 | 12/2018 | Al-Ali et al.   | N/A | N/A |
| 2019/0090748 | 12/2018 | Al-Ali          | N/A | N/A |
| 2019/0090760 | 12/2018 | Kinast et al.   | N/A | N/A |
| 2019/0090764 | 12/2018 | Al-Ali          | N/A | N/A |
| 2019/0117070 | 12/2018 | Muhsin et al.   | N/A | N/A |
| 2019/0117139 | 12/2018 | Al-Ali et al.   | N/A | N/A |
| 2019/0117140 | 12/2018 | Al-Ali et al.   | N/A | N/A |
| 2019/0117141 | 12/2018 | Al-Ali          | N/A | N/A |
| 2019/0117930 | 12/2018 | Al-Ali          | N/A | N/A |
| 2019/0122763 | 12/2018 | Sampath et al.  | N/A | N/A |

|              |         |                      |     |     |
|--------------|---------|----------------------|-----|-----|
| 2019/0239787 | 12/2018 | Pauley et al.        | N/A | N/A |
| 2019/0274627 | 12/2018 | Al-Ali et al.        | N/A | N/A |
| 2019/0320906 | 12/2018 | Olsen                | N/A | N/A |
| 2019/0374713 | 12/2018 | Kiani et al.         | N/A | N/A |
| 2020/0060869 | 12/2019 | Telfort et al.       | N/A | N/A |
| 2020/0111552 | 12/2019 | Ahmed                | N/A | N/A |
| 2020/0113435 | 12/2019 | Muhsin               | N/A | N/A |
| 2020/0113488 | 12/2019 | Al-Ali et al.        | N/A | N/A |
| 2020/0113496 | 12/2019 | Scruggs et al.       | N/A | N/A |
| 2020/0113497 | 12/2019 | Triman et al.        | N/A | N/A |
| 2020/0113520 | 12/2019 | Abdul-Hafiz et al.   | N/A | N/A |
| 2020/0138288 | 12/2019 | Al-Ali et al.        | N/A | N/A |
| 2020/0138368 | 12/2019 | Kiani et al.         | N/A | N/A |
| 2020/0163597 | 12/2019 | Dalvi et al.         | N/A | N/A |
| 2020/0196877 | 12/2019 | Vo et al.            | N/A | N/A |
| 2020/0253474 | 12/2019 | Muhsin et al.        | N/A | N/A |
| 2020/0253544 | 12/2019 | Belur Nagaraj et al. | N/A | N/A |
| 2020/0275841 | 12/2019 | Telfort et al.       | N/A | N/A |
| 2020/0288983 | 12/2019 | Telfort et al.       | N/A | N/A |
| 2020/0321793 | 12/2019 | Al-Ali et al.        | N/A | N/A |
| 2020/0329983 | 12/2019 | Al-Ali et al.        | N/A | N/A |
| 2020/0329984 | 12/2019 | Al-Ali et al.        | N/A | N/A |
| 2020/0329993 | 12/2019 | Al-Ali et al.        | N/A | N/A |
| 2020/0330037 | 12/2019 | Al-Ali et al.        | N/A | N/A |
| 2021/0022628 | 12/2020 | Telfort et al.       | N/A | N/A |
| 2021/0104173 | 12/2020 | Pauley et al.        | N/A | N/A |
| 2021/0113121 | 12/2020 | Diab et al.          | N/A | N/A |
| 2021/0117525 | 12/2020 | Kiani et al.         | N/A | N/A |
| 2021/0118581 | 12/2020 | Kiani et al.         | N/A | N/A |
| 2021/0121582 | 12/2020 | Krishnamani et al.   | N/A | N/A |
| 2021/0161465 | 12/2020 | Barker et al.        | N/A | N/A |
| 2021/0236729 | 12/2020 | Kiani et al.         | N/A | N/A |
| 2021/0256267 | 12/2020 | Ranasinghe et al.    | N/A | N/A |
| 2021/0256835 | 12/2020 | Ranasinghe et al.    | N/A | N/A |
| 2021/0275101 | 12/2020 | Vo et al.            | N/A | N/A |
| 2021/0290060 | 12/2020 | Ahmed                | N/A | N/A |
| 2021/0290072 | 12/2020 | Forrest              | N/A | N/A |
| 2021/0290080 | 12/2020 | Ahmed                | N/A | N/A |
| 2021/0290120 | 12/2020 | Al-Ali               | N/A | N/A |
| 2021/0290177 | 12/2020 | Novak, Jr.           | N/A | N/A |
| 2021/0290184 | 12/2020 | Ahmed                | N/A | N/A |
| 2021/0296008 | 12/2020 | Novak, Jr.           | N/A | N/A |
| 2021/0330228 | 12/2020 | Olsen et al.         | N/A | N/A |
| 2021/0386382 | 12/2020 | Olsen et al.         | N/A | N/A |
| 2021/0402110 | 12/2020 | Pauley et al.        | N/A | N/A |
| 2022/0039707 | 12/2021 | Sharma et al.        | N/A | N/A |
| 2022/0053892 | 12/2021 | Al-Ali et al.        | N/A | N/A |
| 2022/0071562 | 12/2021 | Kiani                | N/A | N/A |
| 2022/0096603 | 12/2021 | Kiani et al.         | N/A | N/A |
| 2022/0151521 | 12/2021 | Krishnamani et al.   | N/A | N/A |

|              |         |                    |     |     |
|--------------|---------|--------------------|-----|-----|
| 2022/0218244 | 12/2021 | Kiani et al.       | N/A | N/A |
| 2022/0287574 | 12/2021 | Telfort et al.     | N/A | N/A |
| 2022/0296161 | 12/2021 | Al-Ali et al.      | N/A | N/A |
| 2022/0361819 | 12/2021 | Al-Ali et al.      | N/A | N/A |
| 2022/0379059 | 12/2021 | Yu et al.          | N/A | N/A |
| 2022/0392610 | 12/2021 | Kiani et al.       | N/A | N/A |
| 2023/0028745 | 12/2022 | Al-Ali             | N/A | N/A |
| 2023/0038389 | 12/2022 | Vo                 | N/A | N/A |
| 2023/0045647 | 12/2022 | Vo                 | N/A | N/A |
| 2023/0058052 | 12/2022 | Al-Ali             | N/A | N/A |
| 2023/0058342 | 12/2022 | Kiani              | N/A | N/A |
| 2023/0087671 | 12/2022 | Telfort et al.     | N/A | N/A |
| 2023/0110152 | 12/2022 | Forrest et al.     | N/A | N/A |
| 2023/0111198 | 12/2022 | Yu et al.          | N/A | N/A |
| 2023/0115397 | 12/2022 | Vo et al.          | N/A | N/A |
| 2023/0116371 | 12/2022 | Mills et al.       | N/A | N/A |
| 2023/0135297 | 12/2022 | Kiani et al.       | N/A | N/A |
| 2023/0138098 | 12/2022 | Telfort et al.     | N/A | N/A |
| 2023/0145155 | 12/2022 | Krishnamani et al. | N/A | N/A |
| 2023/0147750 | 12/2022 | Barker et al.      | N/A | N/A |
| 2023/0210417 | 12/2022 | Al-Ali et al.      | N/A | N/A |
| 2023/0222805 | 12/2022 | Muhsin et al.      | N/A | N/A |
| 2023/0222887 | 12/2022 | Muhsin et al.      | N/A | N/A |
| 2023/0226331 | 12/2022 | Kiani et al.       | N/A | N/A |
| 2023/0284916 | 12/2022 | Telfort            | N/A | N/A |
| 2023/0284943 | 12/2022 | Scruggs et al.     | N/A | N/A |
| 2023/0301562 | 12/2022 | Scruggs et al.     | N/A | N/A |
| 2023/0346993 | 12/2022 | Kiani et al.       | N/A | N/A |
| 2023/0368221 | 12/2022 | Haider             | N/A | N/A |
| 2023/0371893 | 12/2022 | Al-Ali et al.      | N/A | N/A |
| 2023/0389837 | 12/2022 | Krishnamani et al. | N/A | N/A |
| 2024/0016418 | 12/2023 | Devadoss et al.    | N/A | N/A |
| 2024/0016419 | 12/2023 | Devadoss et al.    | N/A | N/A |
| 2024/0047061 | 12/2023 | Al-Ali et al.      | N/A | N/A |
| 2024/0049310 | 12/2023 | Al-Ali et al.      | N/A | N/A |
| 2024/0049986 | 12/2023 | Al-Ali et al.      | N/A | N/A |
| 2024/0081656 | 12/2023 | DeJong et al.      | N/A | N/A |
| 2024/0122486 | 12/2023 | Kiani              | N/A | N/A |
| 2024/0180456 | 12/2023 | Al-Ali             | N/A | N/A |
| 2024/0188872 | 12/2023 | Al-Ali et al.      | N/A | N/A |
| 2024/0245855 | 12/2023 | Vo et al.          | N/A | N/A |
| 2024/0252046 | 12/2023 | Jansen et al.      | N/A | N/A |
| 2024/0260894 | 12/2023 | Olsen              | N/A | N/A |
| 2024/0267698 | 12/2023 | Telfort et al.     | N/A | N/A |
| 2024/0277233 | 12/2023 | Al-Ali             | N/A | N/A |
| 2024/0277280 | 12/2023 | Al-Ali             | N/A | N/A |
| 2024/0298920 | 12/2023 | Fernkbist et al.   | N/A | N/A |
| 2024/0306985 | 12/2023 | Vo et al.          | N/A | N/A |
| 2024/0324953 | 12/2023 | Telfort            | N/A | N/A |
| 2024/0380246 | 12/2023 | Moran              | N/A | N/A |

**FOREIGN PATENT DOCUMENTS**

| Patent No.     | Application Date | Country | CPC |
|----------------|------------------|---------|-----|
| 200010929      | 12/1999          | AU      | N/A |
| 2 346 639      | 12/1999          | CA      | N/A |
| 2 366 493      | 12/2001          | CA      | N/A |
| 0 204 459      | 12/1985          | EP      | N/A |
| 0 313 238      | 12/1988          | EP      | N/A |
| 1 222 894      | 12/2001          | EP      | N/A |
| 1 222 894      | 12/2002          | EP      | N/A |
| 2003-521985    | 12/2002          | JP      | N/A |
| 2004-329406    | 12/2003          | JP      | N/A |
| 2012-0001261   | 12/2011          | KR      | N/A |
| WO 96/031155   | 12/1995          | WO      | N/A |
| WO 00/021433   | 12/1999          | WO      | N/A |
| WO 01/003574   | 12/2000          | WO      | N/A |
| WO 01/041634   | 12/2000          | WO      | N/A |
| WO 02/089664   | 12/2001          | WO      | N/A |
| WO 2010/135373 | 12/2009          | WO      | N/A |

**OTHER PUBLICATIONS**

US 2024/0016391 A1, 01/2024, Lapotko et al. (withdrawn) cited by applicant  
U.S. Pat. No. 8,989,831, Disposable Components for Reusable Physiological Sensor, Mar. 24, 2015. cited by applicant  
U.S. Pat. No. 9,895,107, Disposable Components for Reusable Physiological Sensor, Feb. 20, 2018. cited by applicant  
U.S. Appl. No. 10,342,487, Disposable Components for Reusable Physiological Sensor, Jul. 9, 2019. cited by applicant  
U.S. Appl. No. 16/425,592, Disposable Components for Reusable Physiological Sensor, filed May 29, 2019. cited by applicant  
U.S. Appl. No. 13/571,910, Fingertip Pulse Oximeter, filed Aug. 10, 2012. cited by applicant  
European Examination Report re EPO Application No. 06 83 8888.3, dated Sep. 27, 2011. cited by applicant  
European Extended Search Report re EPO Application No. 11 16 9443.6, dated Sep. 12, 2011. cited by applicant  
International Search Report and Written Opinion issued in PCT application No. PCT/US2010/035323, dated Aug. 13, 2010, in 16 pages. cited by applicant  
International Search Report in PCT application No. PCT/US2006/046176, dated Nov. 29, 2006, in 4 pages. cited by applicant  
Japanese Office Action re JP Application No. 2008-543525, dated Feb. 28, 2012. cited by applicant

---

*Primary Examiner:* Fardanesh; Marjan

*Attorney, Agent or Firm:* Knobbe, Martens, Olson & Bear, LLP

---

**Background/Summary**

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 16/425,592, filed May 29, 2019, which application is a continuation of U.S. patent application Ser. No. 15/864,389, filed Jan. 8, 2018, which application is a continuation of U.S. patent application Ser. No. 14/626,570, filed Feb. 19, 2015, which application is a continuation of U.S. patent application Ser. No. 12/782,651, filed May 18, 2010, which application claims the benefit of priority from U.S. Provisional Application No. 61/179,670, filed May 19, 2009. The entire contents of each of the above items are hereby incorporated by reference herein for all purposes.

## FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to disposable components of non-invasive physiological sensors.

## BACKGROUND OF THE DISCLOSURE

(2) Non-invasive physiological sensors are applied to the body for monitoring or making measurements indicative of a patient's health. One application for a non-invasive physiological sensor is pulse oximetry, which provides a noninvasive procedure for measuring the oxygen status of circulating blood. Oximetry has gained rapid acceptance in a wide variety of medical applications, including surgical wards, intensive care and neonatal units, general wards, and home care and physical training. A pulse oximetry system generally includes a physiological sensor having light emitters and a detector, such as one or more LEDs and a light sensor. The sensor is attached to a tissue site, such as a finger, toe, ear lobe, nose, hand, foot, or other site having pulsatile blood flow which can be penetrated by light from the emitters. The detector is responsive to the emitted light after attenuation by pulsatile blood flowing in the tissue site. The detector outputs a detector signal to a monitor which processes the signal to provide a numerical readout of physiological parameters such as oxygen saturation (SpO<sub>2</sub>) and pulse rate.

(3) High fidelity pulse oximeters capable of reading through motion induced noise are disclosed in U.S. Pat. Nos. 6,770,028, 6,658,276, 6,157,850, 6,002,952, 5,769,785, and 5,758,644, which are assigned to Masimo Corporation ("Masimo") and are incorporated by reference herein. Advanced physiological monitoring systems may incorporate pulse oximetry in addition to advanced features for the calculation and display of other blood parameters, such as carboxyhemoglobin (HbCO), methemoglobin (HbMet) and total hemoglobin (Hbt), total Hematocrit (Hct), oxygen concentrations and glucose concentrations, as a few examples. Advanced physiological monitors and corresponding multiple wavelength optical sensors capable of measuring parameters in addition to SpO<sub>2</sub>, such as HbCO, HbMet and Hbt are described in at least U.S. patent application Ser. No. 11/367,013, filed Mar. 1, 2006, titled Multiple Wavelength Sensor Emitters and U.S. patent application Ser. No. 11/366,208, filed Mar. 1, 2006, titled Noninvasive Multi-Parameter Patient Monitor, assigned to Masimo Laboratories, Inc. and incorporated by reference herein. Further, noninvasive blood parameter monitors and optical sensors including Rainbow™ adhesive and reusable sensors and RAD57™ and Radical-7™ monitors capable of measuring SpO<sub>2</sub>, pulse rate, perfusion index (PI), signal quality (SiQ), pulse variability index (PVI), HbCO and HbMet, among other parameters, are also commercially available from Masimo.

## SUMMARY OF THE DISCLOSURE

(4) Optical sensors are widely used across clinical settings, such as operating rooms, emergency rooms, post anesthesia care units, critical care units, outpatient surgery and physiological labs, to name a few. Use in these settings exposes sensors, and in particular reusable sensors, to the potential risks of contamination and the resulting spread of nosocomial (hospital-acquired) infections. Studies have suggested that visual inspection of sensors may not detect contamination. Further, while a low-level disinfection protocol of alcohol wipes, dilute bleach scrubs or distilled water wipes can be effective when done correctly, reusable sensors may still be at risk for bacteria including MRSA (methicillin resistant *Staphylococcus aureus*). MRSA causes skin infections and

can occasionally spread to almost any other organ in the body, sometimes with life-threatening potential. It is therefore a priority among medical care facilities to prevent contamination of reusable sensors by foreign and infectious materials and to prevent the spread of nosocomial infections. A sterile sensor cover advantageously provides a sterile environment without interfering with sensor functionality and capability. For example, the sterile sensor cover may be substantially impermeable to infectious agents (e.g., bacteria) and substantially optically transparent or transmissive.

(5) Further, sensor elements such as the detector, emitters and associated circuitry can be expensive parts of a patient monitoring system. Often, relatively inexpensive degradable portions of the sensor become damaged due to repeated use or frequent sterilization. For example, the portions of the sensor which contact the user's skin often become soiled or damaged after each use. In such cases, users often disconnect the sensor cable from the monitor and replace the entire sensor. Moreover, in certain cases, such as for portable applications, it is useful for a sensor to be integrated into the patient monitor housing and not be attached by a cable. In such cases the sensor may not be removable from the monitor.

(6) Other attempted solutions that include sensors with flexible, adhesive substrates having disposable and reusable portions. The disposable portion generally includes the adhesive portion of the sensor, which can lose its tack. Such sensors generally wrap around and adhesively attach to the tissue site. However, non-adhesive sensors having rigid housings, such as clip type sensors which clamp onto the tissue of the patient, are also commonly used in pulse oximetry and other patient monitoring applications. These sensors include degradable components that can become damaged or soiled due to frequent use, such as upper and lower pads which contact the user's skin.

(7) It is therefore desirable to decouple the degradable portions from the rest of the sensor so that the degradable portions can be replaced in a cost-effective manner, and the rest of the sensor can be reused. A disposable sensor cartridge according to embodiments of the disclosure is capable of being used with a non-invasive physiological sensor and protects the sensor from damage due to repeated use or sterilization.

(8) In addition, incorrect alignment of a patient's tissue site with the sensor elements can lead to inaccurate results. For example, where the tissue site is a patient's finger, the emitter and detector should generally be aligned with the nail bed of the patient. Often it is difficult to determine whether the sensor is properly aligned because, for example, the sensor housing impedes the view of the tissue site in relation to the emitter and detector. In such cases, it may take the operator a significant amount of time to realize that a sensor is misaligned. As such, there is a need for a sensor which provides for robust tissue site alignment. Embodiments of the sensor cartridge are positionable on the user before placing the treatment site in the sensor, allowing for improved alignment of the treatment site.

(9) In order to provide cost savings and allow for efficient use, it is also advantageous to be able to reuse sensors on different patients having tissue sites of various sizes, such as for both adults and children. The sensor cartridge of certain embodiments can be configured to allow a sensor to comfortably accommodate treatment sites of various sizes, such as for both adult and pediatric applications.

(10) According to certain embodiments, a disposable sensor cartridge is provided for use with a noninvasive physiological sensor. The sensor cartridge can be capable of attaching to a tissue site and mating with a housing of the sensor. In certain embodiments, the sensor cartridge comprises a first portion comprising a first aperture configured to allow light emitted from one or more emitters of the sensor to travel through the first aperture such that the light is incident on a first region of a tissue site and travels through and is attenuated by body tissue of the tissue site. The attenuated light may exit the tissue site at a second region of the tissue site. In certain embodiments, the sensor cartridge also includes a second portion comprising a second aperture configured to allow the attenuated light to travel through the second aperture and to be received by a detector of the sensor.

In some embodiments, the first and second portions coupled to define a cavity capable of receiving the tissue site. In certain embodiments, the sensor cartridge can also include an electrical component capable of electrical communication with the sensor. In some embodiments, the first portion is in contact with the first region while the cartridge is attached to the tissue site and the second portion is in contact with the second region while the cartridge is attached to the tissue site. (11) According to an aspect of the disclosure, a method of using a noninvasive physiological sensor having a housing is provided. The method can include providing a disposable sensor cartridge comprising a first portion comprising a first aperture and a second portion comprising a second aperture. In certain embodiments, the disposable sensor cartridge can also include an electrical component. The first portion and second portion may together define a cavity capable of receiving the tissue site. In certain embodiments, the method further includes attaching the sensor cartridge to the tissue site such that the tissue site is disposed within the cavity. The method according to some embodiments also includes mating the sensor cartridge with the sensor such that the sensor cartridge is disposed within a cavity of the sensor housing. In certain embodiments, mating of the sensor cartridge with the sensor is such that the electrical component is in electrical communication with a portion of the sensor.

(12) In certain embodiments, a disposable sterile barrier is provided. The disposable sterile barrier may interpose a material impermeable to infectious biological substances between a tissue site and surfaces of a sensor configured to grasp the tissue site. The sensor may be configured to transmit optical radiation into the tissue site and to generate a signal responsive to the optical radiation after attenuation by pulsatile blood flow within the tissue site. In certain embodiments, the disposable sterile barrier comprises an optically transparent material that is substantially impermeable to infectious biological substances. The disposable barrier may comprise a first portion of the material formed such that the first portion can be interposed between a tissue site and a first surface of a reusable optical sensor proximate an emitter of the sensor. In certain embodiments, the disposable barrier may comprise a second portion of the material formed such that the second portion can be interposed between the tissue site and a second surface of the sensor proximate a detector of the sensor.

(13) In certain embodiments, an optical sensing method of non-invasively measuring the constituents of pulsatile blood flow within a tissue site without substantial risk nosocomial infection by direct contact between a sensor and the tissue site. The optical sensing method may include providing a reusable optical sensor having an emitter disposed within a first housing and a detector disposed within a second housing, the emitter and detector in communication with a sensor cable, the first and second housings configured to be urged against opposite sides of a tissue site upon application of the sensor to the tissue site, the emitter configured to transmit optical radiation having one or more predetermined wavelengths into the tissue site and the detector configured to receive the optical radiation after attenuation by the tissue site. The sensor may be configured to generate one or more signals indicative of the attenuated radiation, the one or more signals transmitted via the sensor cable to a monitor configured to process the one or more signals to determine one or more physiological parameters of patient. The method may include applying a barrier so as to make physiological measurements without direct contact between the tissue site and the sensor. The barrier can be interposed between the tissue site and the first housing and interposed between the tissue site and the second housing upon application of the tissue site to the sensor, the barrier comprising material substantially impermeable to infectious substances and substantially transparent so as not to substantially distort the physiological measurements.

(14) According to certain aspects of the disclosure, a disposable sterile barrier is provided comprising an elongate tube comprising a cavity. The elongate tube may comprise an open end and a closed end. In certain embodiments, the open end comprises an opening large enough to accommodate a reusable optical sensor. The cavity can be sized to fully enclose the reusable optical sensor and at least a portion of a sensor cable extending from the optical sensor. In some



embodiments, the elongate tube comprises a material which is substantially optically transparent and substantially impermeable to infectious biological substances. In certain embodiments, the disposable sterile barrier may further include a fastener disposed proximate the open end of the elongate tube and configured to seal the open end of the elongate tube around the sensor cable.

---

## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a sensor cartridge and associated sensor according to an embodiment of the disclosure;
- (2) FIGS. 2A-D are side hinged-open, side hinged-closed, top and bottom perspective views, respectively, of the sensor cartridge of FIG. 1;
- (3) FIGS. 3A-D are side hinged-open and side hinged-closed views, top and bottom perspective views, respectively, of a sensor cartridge according to another embodiment of the disclosure;
- (4) FIG. 4 illustrates the mating of the sensor cartridge of FIG. 3 with a sensor.
- (5) FIGS. 5A-B illustrate side views of two sensor cartridges of differing sizes according to an embodiment of the disclosure.
- (6) FIGS. 6A-B illustrate side views of two sensor cartridges of differing sizes according to another embodiment of the disclosure.
- (7) FIG. 7 is a side perspective view of a generally boot-shaped enclosure embodiment of a sterile sensor cover;
- (8) FIG. 8 is a side perspective view of a generally tube-shaped enclosure embodiment of a sterile sensor cover;
- (9) FIGS. 9A-F illustrate enclosing a reusable sensor with the boot-shaped enclosure and sealing the end with an attached tie; and
- (10) FIGS. 10A-E illustrate enclosing a reusable sensor with the tube-shaped enclosure and sealing the end with an adhesive strip.
- (11) FIGS. 11A-B illustrate top front and bottom rear perspective views, respectively, of a sensor cartridge according to an embodiment of the disclosure.
- (12) FIGS. 12A-B illustrate top front and bottom rear perspective views, respectively, of a sensor cartridge according to another embodiment of the disclosure.

### DETAILED DESCRIPTION

(13) A sensor cartridge according to embodiments of the disclosure is capable of being used with a non-invasive physiological sensor. Certain embodiments of the sensor cartridge protect the sensor from damage, such as damage due to repeated use, reduce the need for sensor sanitization, or both. Further, embodiments of the sensor cartridge are positionable on the user before insertion in the sensor and allow for improved alignment of the treatment site with the sensor. In addition, the sensor cartridge of certain embodiments of the disclosure can be configured to allow a single sensor to comfortably accommodate treatment sites of various sizes such as for both adult and pediatric applications. The terms “sensor cover” and “sensor cartridge” are used throughout to describe various embodiments of the disclosure. The terms may be used interchangeably and are not intended to be limiting.

(14) The tissue site of the illustrated embodiments is a finger and the following description therefore refers specifically to the tissue site as a finger for the purposes of clarity. This is not intended to be limiting and, as described herein, the sensor cartridge **110** of certain embodiments may be used with other types of tissue sites.

(15) FIG. 1 is a perspective view of a sensor cartridge **110** and associated sensor **120** according to an embodiment of the disclosure. A user can place a cartridge **110** on a finger of the patient and then insert the finger along with the attached cartridge **110** into a non-invasive physiological sensor

**120.**

(16) The sensor **120** can be a clip-type sensor including an upper housing **122**, a lower housing **124** and a hinge element **126**. The upper and lower housings **122**, **124** house electrical and/or optical components (not shown) of the non-invasive physiological sensor **120**. For example, the upper and lower housings **122**, **124** may house light emitters and a detector of a pulse oximeter sensor, such as one or more LEDs and a light sensor. The sensor **120** can be connected to a patient monitor (not shown) via a cable **128**. For example, the detector outputs a signal to the monitor over the cable **128** which then processes the signal to provide a numerical readout of physiological parameters such as oxygen saturation (SpO<sub>2</sub>) and pulse rate.

(17) After placing the sensor cartridge **110** on the finger, the user can attach the sensor **120** to the patient by applying pressure to the ends **123**, **125** of the upper and lower housings **122**, **124**, forming a cavity **129** capable of receiving the tissue site. Once the patient inserts the tissue site along with the attached sensor cartridge **110** into the cavity **129**, the pressure on the ends **123**, **125** can be released such that the upper and lower housings **122**, **124** come in contact with and secure the tissue site, allowing for accurate non-invasive physiological measurement.

(18) Although disclosed with reference to the sensor of FIG. **1**, an artisan will recognize from the disclosure herein a wide variety of oximeter sensors, optical sensors, noninvasive sensors, medical sensors, or the like that may benefit from the sensor cartridges disclosed herein. The sensor may not be a hinge type sensor but may instead, for example, include an integral housing having an annular ring type opening through which the patient inserts their finger. In various embodiments, the sensor may be adapted to receive a tissue site other than a finger such as a toe, ear lobe, nose, hand, foot, or other site having pulsatile blood flow which can be penetrated by light from the emitter. In addition, the cartridge **110** may be used with a portable monitor and associated sensor components in certain embodiments. Such monitors, including the sensor components, can be integrated into a hand-held device such as a PDA and typically do not include cables or separate monitors. Portable monitors are often used by first responders in emergency situations, in part because of their portability and ease of use. As such, disposable cartridges **110** which can protect the sensor components according to embodiments herein can be of particular benefit when used with spot-check monitors. In addition, in emergency situations medical personnel must treat a large number of patients relatively quickly and it may therefore be difficult to sanitize sensors between patients. Disposable cartridges **110** described herein benefit medical personnel in such cases because a separate pre-sterilized cartridge **110** can be used for each patient.

(19) Referring still to FIG. **1**, the cartridge **110** includes an upper portion **114**, a lower portion **116** and a hinge portion **112** which together form a cavity **118** capable of receiving a finger. The upper portion **114**, lower portion **116** and hinge portion **112** are generally curved so as to accommodate the natural shape of the finger. A hinge cut-out **113** can be disposed on either side of the cartridge **110** near the hinge portion **112**, allowing for the upper and lower portions **114**, **116** to separate from each other by rotating about the hinge portion **112**. To apply the cartridge to the finger, the upper portion **114** and lower portion **116** may be separated such that they rotate about the hinge portion **112** and the cavity **118** becomes large enough to comfortably receive the finger. Once the finger is placed in the cavity **118**, the upper and lower portions **114**, **116** are released, coming into contact with and releasably attaching to the finger. For example, in certain embodiments, the hinge portion **112** may provide a spring force which is transferred to the upper and lower portions **114**, **116** when they are released such that they grab onto the finger. In some embodiments, the interior surface **119** of the cartridge **110** may include an adhesive substance such that the cartridge **110** is releasably attached to the finger or other tissue site. In other configurations, the cartridge **110** simply rests on the finger and there is not a separate attachment mechanism.

(20) The upper portion **114** includes an upper aperture **115** and the lower portion **116** includes a lower aperture **117**. The apertures **115**, **117** generally allow for proper sensor operation. For example, the apertures **115**, **117** allow for light from one or more emitters of the sensor **120** to

contact the finger and for light attenuated by the tissue site to be received by a detector of the sensor **120**.

(21) Because the upper and lower sensor housings **122**, **124** are often opaque, it can be difficult to determine whether or not a finger is properly aligned once it is placed in the sensor. The apertures **115**, **117** can allow for proper alignment of the sensor cartridge **110** with respect to the finger prior to inserting the cartridge **110** and finger into the sensor **120**. Proper alignment of the cartridge **110** with the finger can therefore improve the accuracy of measurements by helping to ensure that the finger will be properly aligned with respect to the sensor **120** elements upon mating of the cartridge **110** and sensor **120**. In certain cases, medical personnel may not realize that an inaccurate measurement has occurred, or may not realize it until after an alarm on the sensor goes off, after removal of the sensor or after the patient has left the treatment facility. As such, proper alignment can also help reduce cost and save time.

(22) The upper aperture **115** allows a user to visually determine whether the nail bed of a finger is properly aligned with the aperture **115**. Proper alignment of the nail bed also ensures that the lower aperture **117** is properly aligned with the fleshy part of the underside of the finger. In addition, the user may visually determine that the underside of the finger is properly aligned by looking through the lower aperture **117**. The finger with the properly aligned cartridge **110** can then be inserted into the sensor **120** such that the light from the emitters of the sensor **120** will be incident on the nail bed through the upper aperture **115**. Further, the detector of the sensor **120** then receives attenuated light from the appropriate portion of the underside of the finger through the lower aperture **117**. In other embodiments, the emitter of the sensor **120** is housed in the lower housing **124** and the detector is housed in the upper housing **122**.

(23) In certain embodiments, a film or other material (not shown) may be included in one or more of the apertures **115**, **117**. For example, in one embodiment, translucent plastic film is placed in the apertures **115**, **117** such that the properties of the emitted and attenuated light passing through the apertures are not affected by the material. In other embodiments, material is used which does affect the optical properties. For example, an optical film which filters out particular wavelengths of light is used in some embodiments.

(24) Tactile feedback elements (not shown) may indicate proper alignment of the cartridge instead of, or in addition to, the apertures **115**, **117**. For example, one or more of the upper and lower portions **114**, **116** may include small protrusions which indicate to the patient whether the cartridge **110** is properly aligned. In one embodiment, for example, protrusions on either side of the finger indicate to the patient that the cartridge **110** is centered on the finger and a protrusion near the hinge portion **112** indicates that the finger is inserted appropriately deep into the cartridge **110**. Other tactile elements such as, for example, recesses, may be used in other embodiments.

(25) In certain embodiments, the properties of the interior **119** of the cartridge **110** allow for proper and efficient calibration of the sensor **120**. For example, the interior **119** of the cartridge may be a highly reflective color such as white. The interior **119** may also have a glossy texture in certain embodiments, which can also aid in the calibration of the sensor.

(26) As discussed, the sensor cartridge **110** is configured to mate with the sensor **120**. For example, the cartridge **110** of FIG. 1 mates in a friction fit with the sensor **120**. To release the cartridge **110** from the sensor **120** after use, the user opens the sensor **120** clip, thereby relieving the pressure on the cartridge and tissue site from the upper and lower housings **122**, **124**. The user can then readily pull the tissue site and cartridge **110** out of the cavity **129**. As will be appreciated by skilled artisans from the disclosure provided herein, other types of mating and release mechanisms are possible. For example, snap-fit mating configurations such as the one described below with respect to FIG. 3 can be used. In some embodiments, a male or female track on the cartridge **110** fits into a corresponding male or female track on the sensor **120**, as appropriate. A button or lever release mechanism may be used in certain embodiments. In some embodiments, the cartridge **110** may automatically release when the sensor **120** clip is opened.

(27) The sensor cartridge **110** or a portion thereof can be constructed of urethane rubber plastic in certain embodiments. In various other configurations, as will be appreciated from the disclosure herein, the cartridge **110** may be made of other appropriate materials such as Acrylonitrile Butadiene Styrene (“ABS”) or other types of rubber or plastic materials. The cartridge **110** is formed with biodegradable material in certain embodiments.

(28) The general structure of the sensor cartridge **110** may differ in various configurations. For example, the cartridge may not be an integral molded piece as shown in the embodiment of FIG. 1 and may instead include separate pieces as discussed with respect to FIG. 3 below.

(29) In an example scenario, once the user attaches the properly aligned sensor cartridge **110** to the tissue site and applies the sensor **120** as discussed above, medical personnel take a measurement of one or more physiological parameter of the patient using the sensor **120**. The patient then removes the tissue site from the sensor **120** along with the attached cartridge **110** as discussed and the patient or medical personnel may dispose of the cartridge **110**. Those of skill in the art will recognize from the disclosure herein various ways of using the cartridge **110**. For example, the cartridge **110** of certain embodiments may be mated with the sensor **120** before application to the tissue site. Moreover, the cartridge may be used more than once. For example, the cartridge **110** may be used several times by the same patient before disposal. In some embodiments, the cartridge **110** may be sterilized between uses.

(30) In certain embodiments, the sensor cartridge **110** is configured to be in electrical communication with the sensor **120**. For example, the sensor cartridge **110** can also include one or more electronic components **130** and one or more coupling portions (not shown) which may be electrically coupled to the sensor **120**. For example, the coupling portions may comprise one or more electrical contacts (e.g., solder pads), electrical connectors (e.g., a socket and pin type connector), and the like. The electronic components **130** may include one or more information elements in certain embodiments. The information element may comprise one or more memory devices, such as, for example, one or more EPROMs or EEPROMs (e.g., those commercially available from Dallas Semiconductor), other memory or processing devices, combinations of the same, or the like. In some embodiments, the information element includes a conductor, a resistor, a single wire addressable memory device, or the like. In general, the information element may include a read-only device or a read and write device. The information element may advantageously comprise a resistor, an active network, or any combination of the foregoing. The information element may include data accessible by the sensor **120** and/or attached patient monitor to accomplish quality control, use monitoring, combinations of the same, or the like. For example, the information element may provide identification information to the system (e.g., the sensor **120** and/or monitor) which the system uses to determine whether the cartridge is compatible with the system. In an advantageous embodiment, the monitor reads the information element to determine one, some or all of a wide variety of data and information, including, for example, a type of patient, cartridge manufacturer information, life data indicating whether the cartridge has been used and/or should be replaced, encryption information, keys, indexes to keys or has functions, or the like monitor or algorithm upgrade instructions or data, some or all of parameter equations, information about the patient, age, sex, medications, and other information that can be useful for the accuracy or alarm settings and sensitivities, trend history, alarm history, sensor life, or the like.

(31) The information element may be positioned on the hinge portion **112**. For example, in one embodiment the information element be embedded in the cartridge **112** material of the hinge portion **112** and is electrically coupled via a connector extending from the hinge portion **112** which mates with a corresponding connector on the interior of the sensor **120**. Skilled artisans will recognize from the disclosure provided herein a variety of configurations for the placement of the information element. For example, in various embodiments, the information element may be located on one or more of the inner surface of the cartridge **110**, the outer surface of the cartridge **110**, or embedded within the cartridge **110** material. Moreover, the information element may be

positioned on the hinge portion **112**, the upper portion **114**, the lower portion **116**, or a combination thereof.

(32) The information element may advantageously store some or all of a wide variety of data and information, including, for example, information on the type or operation of the sensor cartridge, buyer or manufacturer information, software such as scripts, executable code, or the like, sensor cartridge **110** life data indicating whether the sensor cartridge **110** has expired and should be replaced, encryption information, etc. or monitor or algorithm upgrade instructions or data. In various configurations, the information element may advantageously configure or activate the monitor, monitor algorithms, monitor functionality, or the like based on some or all of the foregoing information. For example, without authorized data accessible on the information element, quality control functions may inhibit functionality of the monitor. Likewise, particular data may activate certain functions while keeping others inactive. For example, a particular cartridge **110** may be compatible for use in measuring one type of physiological parameter of a set of physiological parameters the monitor is capable of measuring. In such a circumstance, the monitor may only activate measurement of the one type of physiological parameter based on the data accessible from the information element. Further information regarding information elements and systems and methods for monitoring sensor life can be found in U.S. Publication No. 2008/0088467, which is hereby incorporated in its entirety by reference herein.

(33) While disclosed with respect to the cartridge **110** of FIGS. **1-2**, any of the cartridges described herein, such as the cartridges **300**, **510**, **520**, **610**, **620**, **1100**, **1200** described below of FIGS. **3-6** and FIGS. **11-12** may include one or more information elements. Likewise, any of the sensor covers disclosed herein, such as the sensor covers **700**, **800** of FIGS. **7-10** may include an information element.

(34) FIGS. **2A-D** are side hinged-open, side hinged-closed, top-right perspective and top-left perspective views, respectively, of the sensor cartridge of FIG. **1**. As discussed above, the upper portion **114** and lower portion **116** may be separated such that they rotate about the hinge portion **112** and the cavity **118** becomes large enough to comfortably receive the finger (FIG. **2A**). Once the finger is placed in the cavity **118**, the upper and lower portions **114**, **116** can be released, coming into contact with and releasably attaching to the finger (FIG. **2B**). The right and left sides of the cartridge **110** of the illustrated embodiment are symmetrical. As shown by FIGS. **2C-D**, the top and bottom are also symmetrical. In other embodiments, the left and right sides and/or the top and bottom may be shaped differently, such as, for example, to accommodate asymmetric properties of the finger or other tissue site.

(35) FIGS. **3A-D** are side hinged-open and side hinged-closed views, top-right perspective and top-left perspective views, respectively, of a sensor cartridge **300** according to another embodiment of the disclosure. The cartridge **300** includes an upper portion **314**, a lower portion **316** and a hinge portion **312**, which are three separate pieces. As shown, the hinge portion **312** attaches to the back **302** of the upper portion **314** and the back **304** of the lower portion **316**, forming a cavity **318** capable of receiving a finger of a patient. The hinge portion **312** may attach to the upper and lower portions **312**, **314** with glue, may be formed integrally with the upper and lower portions **312**, **314**, or may be connected in some other manner. As shown with respect to FIGS. **3C-D**, the cartridge **110** further includes upper and lower apertures **315**, **317**, which may be similar in structure and function to the apertures **115**, **117** described above with respect to FIGS. **1-2**.

(36) FIG. **4** illustrates the mating of the sensor cartridge of FIG. **3** with a sensor **420** (shown in a cross-sectional view). The sensor **420** may be similar in structure and function to the sensor **120** of FIG. **1**, for example, and can include an upper housing **422**, a lower housing **424** and a hinge portion **426** which form a cavity **429**. One or more securing features may be included on the cartridge **300** to secure the cartridge **300** with the one or more corresponding securing features on the sensor **420** upon mating to secure the cartridge to the sensor **420** and/or ensure the correct position of the cartridge **300** within the sensor **420**. Referring to FIGS. **3** and **4**, for example, the

attachment arm **311** is configured to secure the cartridge **300** in place when applied to the sensor **420**. Upon application to a sensor **420** the front portion **425** of the lower housing **424** of the sensor **420** may occupy the space defined by the attachment arm **311** and the underside of the lower portion **316** of the cartridge **300**. The attachment arm **311** helps to releasably secure the sensor **420**, via a friction fit, for example. One or more other features, such as the lip **313** disposed on the upper portion **314** of the cartridge **300** can be included to further secure the cartridge **300** in the sensor **420**. Upon insertion of the cartridge **300** into the sensor **420**, the front portion **423** of the upper housing **422** abuts the lip **313**. Accordingly, the lip **313** can help ensure that the cartridge **300** is positioned appropriately deep within the sensor **420**.

(37) The shape of the cartridge **300** may be configured to secure the cartridge **300** and/or ensure proper positioning of the cartridge **300** upon mating with the sensor **420**. For example, as shown in FIGS. 3-4, the upper portion **314** and/or the lower portion **316** can be sloped such that they thicken towards the front **301** of the cartridge **300**. Upon insertion, the sloping feature can provide a friction fit with the corresponding portions of the sensor **420**.

(38) As will be appreciated by skilled artisans from the disclosure provided herein, various attachment and positioning mechanisms may be used. For example, the attachment arm **311** may further include a protrusion or other feature which may fit into a corresponding feature, such as a recess (not shown), on the underside of the lower housing **424** of the sensor **420**. In other configurations, the features may be reversed. For example, the protrusion may be on the lower housing **424** and the recess may be included on the attachment arm **311** of the cartridge **300**.

(39) FIGS. 5A-B illustrate side views of two sensor cartridges **510**, **520** of differing sizes according to embodiments of the disclosure. The sensor cartridges **510**, **520** may be similar in structure and function to the cartridge **100** of FIGS. 1-2 and include upper portions **514**, **524**, lower portions **516**, **526**, hinge portions **512**, **522**, cavities **518**, **528**, upper apertures **515**, **525** and lower apertures **517**, **527**, respectively. The cartridges **510**, **520** are configured to accommodate treatment sites of various sizes, such as for both adult and pediatric applications. For example, the length L.sub.1 of the cartridge **510** of FIG. 5A is larger than the corresponding length L4 of the cartridge **520** of FIG. 5B. Additionally, the upper and lower apertures **515**, **517** of the cartridge **510** of FIG. 5A are set back a length L2 from the rear of the cartridge **510** which is relatively large in comparison to the length L5 that the upper and lower apertures **525**, **527** of cartridge **520** of FIG. 5B are set back from the rear of the cartridge **520**. The length L3 of the upper and lower apertures **515**, **517** of the cartridge **510** of FIG. 5A is relatively large in comparison to the length L6 of the upper and lower apertures **525**, **527** of the cartridge **520** of FIG. 5B.

(40) The cartridge **510** is configured to accommodate a larger finger than the cartridge **520**. For example, the cartridge **510** may be used in adult applications and the cartridge **520** may be used in pediatric applications. A pediatric patient, for example, may have a nail bed that is set back a relatively short distance from the tip of the finger in comparison to an adult patient. As such, the shorter set back length L5 may be appropriate for a pediatric patient while the length L2 may be appropriate for an adult patient. A pediatric patient may also have a relatively smaller nail bed than an adult patient such that the smaller length L6 appropriately accommodates a pediatric patient while the length L3 may more appropriately accommodate an adult patient. Finally, pediatric patients typically have shorter fingers than adult patients such that the shorter overall cartridge length L4 may be appropriate for a pediatric patient while the longer length L.sub.1 may be more appropriate for adult patients.

(41) As shown, the external profile of the cartridges **520**, **521**, characterized in part by the maximum height of the cartridges **520**, **521** H.sub.1 may be the same. As such, the cartridges **520**, **521** may mate with a single sized sensor, thereby reducing cost and complexity. One of skill in the art will recognize from the disclosure herein other configurations. Some of the lengths which are shown as different in the embodiments **510**, **520** of FIGS. 5A-B may be the same in alternative configurations. For example, in various embodiments, one or more of the pairs of lengths L1 and

L4, L2 and L5, L3 and L6 may be the same.

(42) FIGS. 6A-B illustrate side views of two sensor cartridges **610**, **620** of differing sizes according to embodiments of the disclosure. The sensor cartridges **610**, **620** may be similar in structure and function to the cartridge **300** of FIG. 3 and include upper portions **614**, **624**, lower portions **616**, **626**, hinge portions **612**, **622**, cavities **618**, **628**, attachment arms **611**, **621** and lips **613**, **623** respectively. The cartridges **610**, **620** are configured to accommodate treatment sites of various sizes, such as for both adult and pediatric applications. For example, the upper and lower portions **614**, **616** of the cartridge **610** are thinner than the corresponding portions **624**, **626** of the cartridge **620** and define a relatively large cavity **618**, characterized in part by the length H.sub.2. On the other hand, the upper and lower portions **624**, **626** of the cartridge **620** are relatively thick and define a relatively small cavity **628**, characterized in part by the length H.sub.3. As such, the cartridge **610** is configured to accommodate a larger finger than the cartridge **620**. For example, the cartridge **610** may be used in adult applications and the cartridge **620** may be used in pediatric applications.

(43) As shown, the external profile of the cartridges **620**, **621**, characterized in part by the maximum height of the cartridges **620**, **621** H.sub.1 may be the same. As such, the cartridges **620**, **621** may mate with a single sized sensor, thereby reducing cost and complexity. One of skill in the art will recognize from the disclosure herein other configurations. In some embodiments, the hinge portions **612**, **614** are also shaped to accommodate tissue sites of various sizes. In some configurations, the upper portion, the lower portion and/or the hinge portion of the sensor cartridge are constructed of a resilient material (e.g., a spongy material) which conforms to the shape of the finger such that a sensor cartridge of a single size can accommodate tissue sites of multiple sizes. In some embodiments in which the opening of the cartridge is generally circular, the circumference of the opening through which the patient inserts their finger into the cavity **518** is larger than the opening of the cartridge **528**.

(44) FIGS. 7-8 illustrate two embodiments of a sterile cover **700**, **800** each advantageously configured to enclose a reusable sensor **120** (FIG. 1). In this manner, a patient's tissue site, such as a finger, can be inserted into the sensor and a physiological measurement made without contamination of the sensor or the patient. The sensor and the patient are thus advantageously protected from exposure to infectious agents such as MRSA.

(45) As shown in FIG. 7, a boot-shaped cover **700** defines an enclosure having a generally tubular leg **720** and a smaller, generally tubular foot **730** both extending from a cross-member **710**. In some embodiments, the cross-member **710** is sealed or substantially sealed. The leg **720** has an open end **722** of sufficient diameter to receive a reusable sensor **120**, as described with respect to FIGS. 9A-B, below. The foot **730** has a closed end **732** of sufficient diameter to receive the lower housing **124** of a reusable sensor **120**, as described with respect to FIGS. 9C-D, below. A tie **740** wraps around the open end **722** so as to close that end around the sensor cable **128**. In one embodiment, the tie **740** is a self-adhesive tie, for example. In an embodiment, the tie **740** wraps around the open end **722** so as to seal the end around the sensor cable **128**. A gap **750** between the leg **720** and foot **730** of the cover **700** defines a space configured to accommodate a finger-tip within the sensor **120**.

(46) As shown in FIG. 8, a tube-shaped cover **800** defines an enclosure having a generally cylindrical body **810** with an open end **812** and a closed end **814**. The open end **812** is of sufficient diameter to receive first the lower housing **124** followed by the upper housing **122** of a reusable sensor **120**, as described with respect to FIGS. 10A-D, below. A self-adhesive strip **820** disposed internal to the body **810** proximate the open end **812** closes the open end **812** around the sensor cable **128**. In one embodiment, for example, the open end **812** forms a seal around the sensor cable **128**. In an embodiment, the covers **700**, **800** include of an optically transparent or substantially transparent material so as to negligibly attenuate or otherwise negligibly distort those wavelengths in the red and infrared spectrum utilized in pulse oximeters or otherwise for the measurement of blood constituents. For example, the covers **700**, **800** may be substantially optically transmissive.

(47) As used herein, the term “tube” is used to describe a generally elongate member comprising a cavity and may include an open end and a closed end, for example. As used herein, a tube may have a variety of shapes and characteristics. For example, the tube **800** of various may comprise a cylindrical, rectangular, ovular, triangular, or some other cross-sectional shape, or may be generally deformable.

(48) FIGS. **9A-F** illustrate the use of a sterile cover **700** embodiment to enclose a reusable sensor **120**, thereby advantageously protecting the sensor from contamination and the patient from exposure to infectious agents. A reusable sensor **120** in its normally closed position is initially inserted into the open end **722** of the cover **700** (FIG. **9A**). The sensor **120** is then pushed through the length of the leg **720** until, for example, the sensor abuts the wall of the cross-member **710** (FIG. **9B**). The sensor grips **123**, **125** are pressed from outside the cover **700** so as to move the sensor **120** to its open position (FIG. **9C**). While maintaining pressure on the grips **123**, **125**, the sensor lower housing **124** slides into the foot **730** until it abuts the closed end **732** (FIG. **9D**). Pressure is released from the grips **123**, **125** so that the sensor **120** returns to its closed position over the gap **750** (FIG. **9E**). The tie **740** is wrapped around the leg **720** so as to close-off the end **722** tightly around the sensor cable **128** (FIG. **9F**). The covered sensor **120** is now ready for single-patient use by squeezing the grips **123**, **125** to open the sensor and by inserting a patient finger into the cover gap **750**, which is now disposed inside the sensor **120**, between the emitters and detector.

(49) FIGS. **10A-E** illustrate the use of another sterile cover **800** embodiment to enclose a reusable sensor **120**, likewise substantially protecting the sensor and patient from contamination and/or infectious agents. Sensor grips **123**, **125** are pressed to place the sensor **120** in an open position (FIG. **10A**). The cover **800** may be initially in an “accordion” state or otherwise folded, rolled or compressed, or otherwise collapsible to a reduced overall length. The cover opening **812** is slipped over the lower housing **124** (FIG. **10A**). Maintaining the sensor **120** in the open position, the cover **800** is unfolded or otherwise extended in length as the opening **812** is pulled over the grips **123**, **125** and onto the upper housing **122** (FIG. **10B**). The cover opening **812** is pulled further along the upper housing **122** and onto the sensor cable **128** (FIGS. **10C-D**). The sensor grips **123**, **125** are released so that the sensor **120** returns to its normally closed position (FIG. **10E**). The self-adhesive strip **820** just inside the opening **812** is then exposed, such as by peeling back and removing a liner, and the opening **812** is securely closed around the cable **128** (FIG. **10E**). The covered sensor **120** is now ready for single-patient use by squeezing the grips **123**, **125** to open the sensor and by inserting **850** a patient finger into the sensor between the folded over wall **810** portions of the cover **800** (FIG. **10E**). This places the patient's fingertip inside the sensor **120**, between the emitters and detector.

(50) Skilled artisans will recognize a variety of alternatives sensor covers **700**, **800** from the disclosure provided herein. For example, in other embodiments, the covers **700**, **800** may include one or more apertures such as one or more of the apertures described herein. The covers **700**, **800** may comprise a rigid or semi-rigid material. Moreover, the covers **700**, **800** may be used in addition to a cartridge such as one or more of the cartridges described herein with respect to FIGS. **1-6** and **11-12**. In one alternative embodiment, the cover **700**, **800** may extend beyond the sensor **120** along some length of the cable **128**. For example, the cover **700**, **800** may extend substantially along the entire cable to the monitor in one embodiment. In another embodiment, the cover **700**, **800** may extend partially along the cable **128**.

(51) FIGS. **11A-B** illustrate top front and bottom rear perspective views, respectively, of a sensor cartridge **1100** according to an embodiment of the disclosure. In certain circumstances, light incident on the finger from the emitter may not be entirely absorbed by the finger. In addition, a certain portion of attenuated light will exit the other side of the finger. Such portions of light may reflect within the space between the cartridge and the finger or between the cartridge and the sensor. These portions of light can interfere with each other, with the emitted light and/or with the attenuated light. These portions of reflected light and associated interfering light can be incident on



the detector. This phenomena, sometimes referred to herein as “light bounce,” can potentially cause inaccurate measurements, depending on certain factors such as the amount of light bounce and the relative sensitivity of the measured signal.

(52) The sensor cartridge **1100** is configured for application to a tissue site such as a finger, for example. The finger or other tissue site may be placed in the opening **1102**. In addition, the cartridge **1100** is configured for insertion into a reusable sensor such as the sensor **120**. The sensor cartridge **1100** further includes an upper aperture **1104** and a lower aperture **1106**. The apertures **1104**, **1106** generally allow for proper sensor operation. For example, the apertures **1104**, **1106** allow for light from one or more emitters of the sensor **120** to contact the finger and for light attenuated by the tissue site to be received by a detector of the sensor **120**. The apertures **1104**, **1106** may function in a manner similar to and provide similar advantages as the apertures **115**, **117** of the cartridge **110** of FIGS. 1-2. The cartridge may comprise Acrylonitrile Butadiene Styrene (“ABS”), other types of rubber or plastic materials, or some other material compatible with the embodiments described herein.

(53) As shown, the sensor cartridge **1100** generally envelopes the finger when it is applied to the cartridge **1100**. Moreover, the cartridge **1100** may comprise a color absorptive of the light emitted from the emitter of the sensor **120**. For example, the cartridge **1100** may comprise a dark material as shown, such as a substantially black or opaque material. As described above, a certain portion of incident on the finger from the emitter of the sensor **120** may not be absorbed by the finger, but may instead be reflected off of the finger. In addition, a certain portion of attenuated light will exit the other side of the finger. Because the cartridge **1100** generally envelopes the finger and is absorptive of the emitted light, light which is not absorbed by the finger or otherwise may advantageously escape into the region between the cartridge and the finger will be substantially absorbed by the cartridge **1100** due to its absorptive properties, reducing the effect of light bounce.

(54) Alternative configurations of the cartridge **1100** are possible as will be recognized by skilled artisans from the disclosure herein. For example, the cartridge **1100** body may comprise a substantially thin rubber material in some embodiments. The cartridge **1100** may include an adhesive proximate the perimeter of the opening **1102**, such as the self-adhesive strip **820** of the cartridge **800** of FIG. 8, which may help create a seal between the finger and the cartridge **1100**. In other configurations, the perimeter of the opening may include an elastic band capable of creating an elastic seal between the finger and the cartridge **1100**. In certain embodiments, the cartridge **1100** or a portion thereof may include a different color such as a relatively light color, or may be translucent or partially translucent. In one configuration, the interior of the cartridge **1100** comprises a substantially dark color and the outer surface comprises another color, such as a light color. In yet another embodiment, the interior of the cartridge **110** comprises a substantially light color and the outer surface comprises a substantially dark color.

(55) FIGS. 12A-B illustrate top front and bottom rear perspective views, respectively, of a sensor cartridge according to another embodiment of the disclosure. The cartridge **1200** of FIG. 12 may be generally similar in structure and function to the cartridge **1100** of FIG. 11. For example, the cartridge **1200** includes an opening **1202**, a cavity **1208**, an upper aperture **1204** and a lower aperture **1206**. In addition, the cartridge **1200** comprises a color absorptive of the emitted light and advantageously reduces the effect of light bounce.

(56) The cartridge **1200** may not fit in a uniformly snug manner with the upper and lower sensor housings **122**, **124**. For example, as the upper and lower sensor housings **122**, **124** exert force directly on the outer surface of the cartridge **1200** and thus indirectly on the tissue site within the cartridge **1200**, the sensor cartridge **1200** may flex and partially deform in response to this force. This flexing may create gaps between portions of the inner surfaces of the upper and lower sensor housings **122**, **124** and the outer surface of the cartridge **1200**. The cartridge **1200** includes connecting portions **1210**, **1212** which are configured to bridge potential gaps between the outer surface of the cartridge **1200** and the inner surfaces of the upper and lower sensor housings **122**,

**124.** Particularly, the connecting portions **1210**, **1212** are configured to bridge these gaps substantially in the region of the perimeter of the apertures **1204**, **1206**, respectively. The connecting portions **1210**, **1212** thereby create a seal between the perimeter of the apertures **1204**, **1206** and the inner surfaces of the upper and lower sensor housings **122**, **124**, respectively. Channel regions are thus created whereby light from the emitter can travel from the emitter to the aperture **1204** and whereby attenuated light can travel from the tissue site to the detector. Thus, the connecting portion **1210** prevents light directed towards the upper aperture **1204** from the emitter from escaping into the region between the inner surface of the upper sensor housing **122** and the outer surface of the cartridge **1200**. Likewise, the connecting portion **1212** prevents attenuated light exiting the tissue site towards the detector of the sensor **120** from prevented from escaping into the region between the inner surface of the lower sensor housing **124** and the cartridge **1200**. Because light escaping into these regions may contribute to light bounce, as described above, the connecting portions **1210**, **1212** advantageously provide further reduction of light bounce. Moreover, the connecting portions **1210**, **1212** generally cause a greater percentage of light from the emitter to be directly incident on the tissue site and a cause greater percentage of attenuated light exiting from the tissue site to be directly incident on the detector. The connecting portions **1210**, **1212** thus can provide for increased measurement accuracy, improved calibration and efficiency of sensor operation, among other advantages.

(57) The connecting portions **1210**, **1212** of the illustrated embodiment include four panels each extending from one side of the perimeter of the upper and lower apertures **1204**, **1206** to form raised rectangular borders around the apertures **1204**, **1206**. The connecting portions **1210**, **1212** are configured to contact the interior surfaces of the upper and lower sensor housings **122**, **124** around the emitter and the detector, respectively. The connecting portions **1210**, **1212** include a flexible material such as a rubber or plastic which conforms to the interior surfaces of the upper and lower sensor housings **122**, **124**, respectively. As such, the connecting portion **1210** helps to create a seal between the emitter of the sensor **120** and the cartridge as described above, thereby reducing light bounce and providing for increased measurement accuracy. Likewise, the connecting portion **1212** helps to create a seal between the detector of the sensor **120** and the cartridge **1200** as described above, thereby further reducing light bounce and improving measurement accuracy.

(58) As will be appreciated by those of skill in the art from the disclosure provided herein, alternative configurations of the cartridge **1200** are possible. For example, in certain embodiments, one or more of the connecting portions **1210**, **1212** may comprise a rigid material. The connecting portions **1210**, **1212** may mate with corresponding features of the interior surfaces of the upper and lower sensor housings **122**, **124**, respectively. For example, the connecting portions **1210**, **1212** may form raised features such as in the illustrated embodiment which fit into corresponding female portions, such as recesses in the interior surfaces of the sensor housings **122**, **124**. In various embodiments, snap-fit, friction-fit, and other mating mechanisms may be employed. Certain features may be reversed in some embodiments. For example, one or more of the interior surfaces of the sensor housings **122**, **124** may include male portions and the connecting portions **1210**, **1212** may include female portions. In certain embodiments, connecting portions such as the connecting portions **1210**, **1212** may be used on other cartridges described herein, such as, for example, the cartridge **110** of FIGS. 1-2, or the cartridge **300** of FIG. 3.

(59) Various sensor cartridges and covers have been disclosed in detail in connection with various embodiments. These embodiments are disclosed by way of examples only and are not to limit the scope of the claims that follow. One of ordinary skill in the art will appreciate the many variations, modifications and combinations. For example, in one embodiment, the cartridge **110** of FIG. 2 includes a securing feature such as the securing feature **311** of the cartridge **300** of FIG. 3. In some embodiments, the cartridges **110**, **300** of FIGS. 2 and 3 are configured to substantially envelope the tissue site in a manner similar to the cartridges **1100**, **1200** of FIGS. 11 and 12. In various embodiments, any of the cartridges described throughout the disclosure, such as the cartridges **300**,

510, 610, 1100, 1200 described with respect to FIGS. 3, 5, 6, 11 and 12, may include a film covering the respective apertures. For example, the film may include a film such as the one described above with respect to the apertures 115, 117 of the cartridge 110 of FIGS. 1-2. In addition, in various embodiments the sensor cartridges and covers are used with a sensor that may measure any type of physiological parameter. In various embodiments, the sensor cartridges and covers may be for any type of medical device.

## Claims

1. A disposable sterile barrier configured to interpose a material between a tissue site and surfaces of a sensor configured to grasp the tissue site, the sensor configured to transmit optical radiation into the tissue site and to generate a signal responsive to the optical radiation after attenuation by pulsatile blood flow within the tissue site, the disposable sterile barrier comprising: a barrier comprising a collapsible tube the barrier being made of a substantially optically transparent material; a first portion of the collapsible tube including an open end, wherein the first portion is sized to receive a reusable optical sensor in the collapsible tube, and wherein at least a portion of the first portion is configured to be interposed between a tissue site and a first surface of the reusable optical sensor proximate an emitter of the reusable optical sensor; and a second portion of the collapsible tube including a closed end, wherein the second portion is sized to receive the reusable optical sensor in the collapsible tube, and wherein at least a portion of the second portion is configured to be interposed between the tissue site and a second surface of the reusable optical sensor proximate a detector of the reusable optical sensor.
2. The disposable sterile barrier of claim 1, wherein: the barrier comprising the collapsible tube is configured to be placed over the reusable optical sensor and over at least a portion of a sensor cable extending from the reusable optical sensor, and the barrier further comprises a fastener proximate the open end of the collapsible tube, the fastener configured to close the open end of the collapsible tube around the sensor cable.
3. The disposable sterile barrier according to claim 2, wherein the collapsible tube is capable of accommodating a first sensor housing comprising one of the first and second surfaces and to accommodate a second sensor housing comprising the other of the first and second surfaces so as to enclose the reusable optical sensor within the collapsible tube.
4. The disposable sterile barrier of claim 1, wherein the tissue site is at least a portion of a finger.
5. An optical sensing method of non-invasively measuring constituents of pulsatile blood flow within a tissue site without substantial risk of nosocomial infection by direct contact between a reusable optical sensor and the tissue site, the optical sensing method comprising: providing a reusable optical sensor having an emitter disposed within a first housing and a detector disposed within a second housing, the emitter and detector in communication with a sensor cable, the first and second housings configured to be urged against opposite sides of a tissue site upon application of the reusable optical sensor to the tissue site, the emitter configured to transmit optical radiation having one or more predetermined wavelengths into the tissue site and the detector configured to receive the optical radiation after attenuation by the tissue site, the reusable optical sensor configured to generate one or more signals indicative of the attenuated radiation, the one or more signals transmitted via the sensor cable to a monitor configured to process the one or more signals to determine one or more physiological parameters of a patient; and applying a barrier so as to make physiological measurements without direct contact between the tissue site and the reusable optical sensor, the barrier configured to be interposed between the tissue site and the first housing and configured to be interposed between the tissue site and the second housing upon application of the tissue site to the reusable optical sensor, the barrier comprising material substantially transparent so as not to substantially distort the physiological measurements, the barrier further comprising a collapsible tube comprising a first portion and a second portion, the first portion

comprising an open end and sized to receive the reusable optical sensor, the second portion comprising a closed end and sized to receive the reusable optical sensor.

6. The optical sensing method of claim 5, further comprising disposing of the barrier after determining one or more physiological parameters of the patient.

7. The optical sensing method of claim 5, wherein applying further comprises: providing the barrier comprising the collapsible tube; expanding the collapsible tube from a collapsed state to an un-collapsed state; fitting the open end of the collapsible tube around one of the first sensor housing and the second sensor housing; moving the open end firstly along one of the first and second housings and then along the other of the first and second housings such that the reusable optical sensor and at least a portion of the sensor cable are enclosed within the tube; and sealing the open end around the sensor cable.

8. The optical sensing method of claim 5, wherein the tissue site is at least a portion of a finger.

9. A disposable sterile barrier comprising: an elongate tube comprising a cavity, an open end and a closed end, the open end comprising an opening large enough to accommodate a reusable optical sensor, the cavity sized to fully enclose the reusable optical sensor and at least a portion of a sensor cable extending from the optical sensor, the elongate tube comprising a material which is substantially optically transparent; and a fastener disposed proximate the open end of the elongate tube and configured to seal the open end of the elongate tube around the sensor cable.

---